Statistical Foundations of Learning

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1 Statistical learning problem and Bayes risk

1.1 Learning problem and risk

Setup. Inputs live in a **feature space** $\mathcal{X}$, outputs in a **label set** $\mathcal{Y}$ (binary classification: $\mathcal{Y} = \{0, 1\}$ or $\{\pm 1\}$; regression: $\mathcal{Y} \subseteq \mathbb{R}$). A **predictor** is a map $h : \mathcal{X} \to \mathcal{Y}$. Given a **training sample** $S = ((x_i, y_i))_{i=1}^m \in (\mathcal{X} \times \mathcal{Y})^m$, a **learner** $\mathcal{A}$ returns a predictor $\hat{h}_S = \mathcal{A}(S)$; it must accept samples of every size $m \geq 1$ (think of a neural network trained by gradient descent on S).

Data model. Learning only makes sense if test data resemble training data. We assume a fixed but unknown distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$: training and test pairs are all drawn from $\mathcal{D}$, independently, so $S \sim \mathcal{D}^m$. Equivalently, $\mathcal{D}$ is specified by the marginal $\mathcal{D}_\mathcal{X}$ of x and the conditional law $\mathbb{P}(y \mid x)$ of the label. A deterministic labelling $y = f(x)$ is the special case of a degenerate conditional; it cannot express that two equal inputs may carry different labels.

Loss and risk. A **loss** $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, \infty)$ penalises a prediction $h(x)$ when the truth is y. The two standard choices are the **0-1 loss** $\mathbb{1}\{h(x) \neq y\}$ for classification and the **squared loss** $(h(x) - y)^2$ for regression.

Definition 1.1 (Risk)**.** The **risk** (generalisation error) of h is its average loss on a new pair from $\mathcal{D}$,

$$L_\mathcal{D}(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}\big[\ell(h(x), y)\big].$$

For a learner, the quantity of interest is $L_\mathcal{D}(\hat{h}_S)$, the expected test error of the predictor learned from S.

Under the 0-1 loss, $L_\mathcal{D}(h)$ is the probability that h misclassifies, $L_\mathcal{D}(h) = \mathbb{P}_{(x,y)\sim\mathcal{D}}(h(x) \neq y)$.

Example 1.2 (Two Gaussian classes)**.** Take $\mathcal{X} = \mathbb{R}$, $\mathcal{Y} = \{\pm 1\}$, both labels with probability $\frac{1}{2}$, and class-conditional laws $\mathcal{N}(-\mu, \sigma^2)$ for $y = -1$ and $\mathcal{N}(\mu, \sigma^2)$ for $y = +1$, with $\mu > 0$. Then $\mathcal{D}_\mathcal{X} = \frac{1}{2}\mathcal{N}(-\mu, \sigma^2) + \frac{1}{2}\mathcal{N}(\mu, \sigma^2)$, and $\mathbb{P}(y = +1 \mid x)$ increases from 0 to 1, crossing $\frac{1}{2}$ at $x = 0$. A threshold rule $h(x) = \operatorname{sign}(x - t)$ errs on the $+1$ points left of t and on the -1 points right of t (fig. 1); moving t away from 0 trades errors on one class for errors on the other. The optimal threshold is $t = 0$, where $\operatorname{sign}(x)$ always predicts the more likely label (theorem 1.6); in practice it has to be found from S alone.

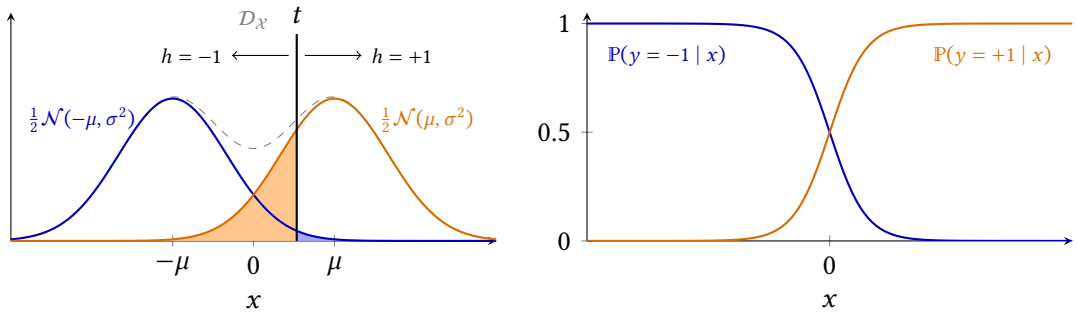


Figure 1: Left: weighted class densities, $\mathcal{D}_\mathcal{X}$ (dashed), errors of $\operatorname{sign}(x - t)$ (shaded). Right: $\mathbb{P}(y = \pm 1 \mid x)$.

1.2 Empirical risk minimisation

Ideally we would solve $\min_{h \in \mathcal{Y}^\mathcal{X}} L_\mathcal{D}(h)$, but $L_\mathcal{D}(h)$ cannot be evaluated without knowing $\mathcal{D}$, so it is replaced by its sample counterpart.

Definition 1.3 (Empirical risk, ERM)**.** The **empirical risk** (training error) of h on S is $L_S(h) = \frac{1}{m}\sum_{i=1}^m \ell(h(x_i), y_i)$. **Empirical risk minimisation** (ERM) solves $\min_{h \in \mathcal{Y}^\mathcal{X}} L_S(h)$.

For fixed h, $L_S(h)$ is an unbiased estimate of $L_D(h)$: each $(x_i, y_i) \sim D$, so by linearity $\mathbb{E}_{S\sim D^m}[L_S(h)] = \frac{1}{m}\sum_{i=1}^m \mathbb{E}\big[\ell(h(x_i), y_i)\big] = L_D(h)$. Minimising it over *all* functions, however, can overfit badly:

Example 1.4 (ERM can fail)**.** Let $x \sim \text{Uniform}([0,1])$ and $y = \mathbb{1}\{x > 0.1\}$. The predictor that outputs 1 on the training points labelled 1 and 0 everywhere else has $L_S(\hat{h}) = 0$. A new x falls in the finite set $\{x_1, \dots, x_m\}$ with probability 0, so $\hat{h}(x) = 0$ almost surely and $\hat{h}$ errs exactly when $y = 1$: $L_D(\hat{h}) = \mathbb{P}(y = 1) = \mathbb{P}(x > 0.1) = 0.9$.

1.3 Bayes risk

Suppose now that D is known. The **Bayes risk** is $L_D^* = \min_{h\in\mathcal{Y}^{\mathcal{X}}} L_D(h)$, and a minimiser h^* is a **Bayes predictor**.

Classification. Let $\mathcal{Y} = \{0,1\}$, ℓ the 0-1 loss, and write $\eta(x) = \mathbb{P}(y = 1 \mid x)$, so that $\mathbb{P}(y = 0 \mid x) = 1 - \eta(x)$.

Theorem 1.5 (Risk of a binary classifier)**.** *For every deterministic $h : \mathcal{X} \to \{0,1\}$,*

$$L_D(h) = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\mathbb{P}(h(x) \neq y \mid x)\big] = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\eta(x)\,\mathbb{1}\{h(x) = 0\} + (1 - \eta(x))\,\mathbb{1}\{h(x) = 1\}\big].$$

Proof. By the tower property,

$$L_D(h) = \mathbb{E}_{(x,y)}\big[\mathbb{1}\{h(x) \neq y\}\big] = \mathbb{E}_x\Big[\mathbb{E}_{y|x}\big[\mathbb{1}\{h(x) \neq y\}\big]\Big] = \mathbb{E}_x\big[\mathbb{P}(h(x) \neq y \mid x)\big].$$

An error occurs either when $h(x) = 0,\ y = 1$ or when $h(x) = 1,\ y = 0$, so $\mathbb{1}\{h(x) \neq y\} = \mathbb{1}\{h(x) = 0\}\mathbb{1}\{y = 1\} + \mathbb{1}\{h(x) = 1\}\mathbb{1}\{y = 0\}$. Given x, $h(x)$ is a constant (h is deterministic), so only the indicators of y are averaged, with $\mathbb{E}_{y|x}[\mathbb{1}\{y = 1\}] = \eta(x)$ and $\mathbb{E}_{y|x}[\mathbb{1}\{y = 0\}] = 1 - \eta(x)$:

$$\mathbb{P}(h(x) \neq y \mid x) = \eta(x)\,\mathbb{1}\{h(x) = 0\} + (1 - \eta(x))\,\mathbb{1}\{h(x) = 1\}. \qquad \square$$

At each x, a classifier therefore pays $\eta(x)$ by predicting 0 or $1 - \eta(x)$ by predicting 1. The best choice is the cheaper option, which defines the **Bayes classifier**

$$h^*(x) = \mathbb{1}\{\eta(x) \geq \tfrac{1}{2}\}.$$

Theorem 1.6 (Bayes risk equals the risk of the Bayes classifier)**.** *For the 0-1 loss,*

$$L_D^* = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\min\{\eta(x), 1 - \eta(x)\}\big] = L_D(h^*) = \min_{h\in\{0,1\}^{\mathcal{X}}} L_D(h).$$

Proof. Fix x. Exactly one of $\mathbb{1}\{h(x) = 0\}$, $\mathbb{1}\{h(x) = 1\}$ equals 1, so the integrand of theorem 1.5 is either $\eta(x)$ or $1-\eta(x)$, hence $\geq \min\{\eta(x), 1-\eta(x)\}$. Taking $\mathbb{E}_x$, every h satisfies $L_D(h) \geq \mathbb{E}_x[\min\{\eta(x), 1-\eta(x)\}]$. For h^*: if $\eta(x) \geq \frac{1}{2}$ then $h^*(x) = 1$ and the integrand is $1 - \eta(x) = \min\{\eta(x), 1 - \eta(x)\}$; if $\eta(x) < \frac{1}{2}$ then $h^*(x) = 0$ and it is $\eta(x) = \min\{\eta(x), 1 - \eta(x)\}$. So h^* attains the lower bound, which is therefore the minimum. $\square$

Example 1.7 (Installation crash)**.** Let $x \sim \text{Uniform}[0,1]$ be the completed fraction of a software installation, with $y = 1$ meaning that the system crashes after the fraction x has been completed, and $\eta(x) = |1 - 2x|$. Then $\eta(x) \geq \frac{1}{2}$ iff $x \leq \frac{1}{4}$ or $x \geq \frac{3}{4}$, so $h^* = 1$ on $[0, \frac{1}{4}] \cup [\frac{3}{4}, 1]$ and $h^* = 0$ on $(\frac{1}{4}, \frac{3}{4})$. The function $\min\{\eta, 1 - \eta\}$ is symmetric about $x = \frac{1}{2}$, and on $[0, \frac{1}{2}]$ (where $\eta = 1 - 2x$, $1 - \eta = 2x$) it equals $2x$ for $x \leq \frac{1}{4}$ and $1 - 2x$ for $x \geq \frac{1}{4}$. Hence

$$L_D^* = L_D(h^*) = 2\Big(\int_0^{1/4} 2x\,dx + \int_{1/4}^{1/2} (1 - 2x)\,dx\Big) = 2\Big(\frac{1}{16} + \frac{1}{16}\Big) = \frac{1}{4}.$$

Regression. Let $\mathcal{Y} = \mathbb{R}$ with the squared loss, so that $L_D(h) = \mathbb{E}[(h(x) - y)^2]$. ERM is then least-squares regression (ordinary least squares when h is restricted to linear functions).

Proposition 1.8 (Bayes regressor)**.** *For the squared loss, $h^*(x) = \mathbb{E}[y \mid x]$ and $L_D^* = \mathbb{E}_x\big[\operatorname{Var}(y \mid x)\big]$.*

Proof. Fix x; for deterministic h, $h(x)$ is a constant given x. Expanding the square and then completing it,

$$\begin{aligned}\mathbb{E}\big[(h(x) - y)^2 \mid x\big] &= h(x)^2 - 2h(x)\,\mathbb{E}[y \mid x] + \mathbb{E}[y^2 \mid x] \\ &= \big(h(x) - \mathbb{E}[y \mid x]\big)^2 + \underbrace{\mathbb{E}[y^2 \mid x] - (\mathbb{E}[y \mid x])^2}_{=\operatorname{Var}(y|x)}.\end{aligned}$$

Taking $\mathbb{E}_x$ gives $L_D(h) = \mathbb{E}_x\big[(h(x) - \mathbb{E}[y \mid x])^2\big] + \mathbb{E}_x[\operatorname{Var}(y \mid x)]$. The second term does not depend on h; the first is ≥ 0 and equals 0 for $h(x) = \mathbb{E}[y \mid x]$. □

Example 1.9 (Signal plus noise)**.** If $y = f(x) + \epsilon$ with ϵ independent of x, $\mathbb{E}[\epsilon] = 0$ and $\mathbb{E}[\epsilon^2] = \sigma^2$, then

$$\mathbb{E}[y \mid x] = f(x) + \mathbb{E}[\epsilon] = f(x), \qquad \operatorname{Var}(y \mid x) = \operatorname{Var}(\epsilon) = \mathbb{E}[\epsilon^2] - (\mathbb{E}[\epsilon])^2 = \sigma^2,$$

so by proposition 1.8 $h^* = f$ and $L_D^* = \sigma^2$.

2 Risk of linear regression

2.1 Least squares and the linear model

Ordinary least squares. Let $\mathcal{X} = \mathbb{R}^p$, $\mathcal{Y} = \mathbb{R}$ and $S = ((x_i, y_i))_{i=1}^m \subset \mathbb{R}^p \times \mathbb{R}$. A **linear model** predicts $h_w(x) = x^\top w$ with $w \in \mathbb{R}^p$. The **ordinary least squares** (OLS) estimator is ERM with the squared loss over linear models,

$$\hat{w} = \operatorname*{arg\,min}_{w \in \mathbb{R}^p} L_S(h_w), \qquad L_S(h_w) = \frac{1}{m}\sum_{i=1}^m \big(y_i - x_i^\top w\big)^2.$$

Writing $\hat{\Sigma} = \frac{1}{m}\sum_{i=1}^m x_i x_i^\top$ (a symmetric positive semidefinite $p \times p$ matrix), the minimiser is, whenever $\hat{\Sigma}$ is invertible,

$$\hat{w} = \hat{\Sigma}^{-1}\Big(\frac{1}{m}\sum_{i=1}^m y_i x_i\Big).$$

Ridge regression. When $\hat{\Sigma}$ is singular, the **ridge** estimator with parameter $\lambda > 0$ replaces it by $\hat{\Sigma} + \lambda I$, whose eigenvalues are all $\geq \lambda$, so that it is always invertible. Equivalently, ridge is ERM penalised by the squared Euclidean norm of w:

$$\hat{w}_\lambda = \big(\hat{\Sigma} + \lambda I\big)^{-1}\Big(\frac{1}{m}\sum_{i=1}^m y_i x_i\Big) = \operatorname*{arg\,min}_{w \in \mathbb{R}^p}\ \frac{1}{m}\sum_{i=1}^m \big(y_i - x_i^\top w\big)^2 + \lambda\|w\|^2.$$

Data model. Since y is continuous, its conditional law given x is specified through a model rather than through the probabilities of single values. Let $\mathcal{D}_\mathcal{X}$ be a distribution on $\mathbb{R}^p$ with $\Sigma = \mathbb{E}_x[xx^\top]$, fix $w^* \in \mathbb{R}^p$, and set

$$y = x^\top w^* + \epsilon, \qquad \epsilon \text{ independent of } x, \quad \mathbb{E}[\epsilon] = 0, \quad \mathbb{E}[\epsilon^2] = \sigma^2.$$

Then $\mathbb{E}[y \mid x] = x^\top w^*$ and $\operatorname{Var}(y \mid x) = \sigma^2$, so by proposition 1.8 the Bayes predictor is h_{w^*} and $L_D^* = \sigma^2$. For $S \sim \mathcal{D}^m$ we write $y_i = x_i^\top w^* + \epsilon_i$, where $\epsilon_1, \dots, \epsilon_m$ are i.i.d. copies of ϵ, independent of $x_1, \dots, x_m$. The estimator $\hat{w}$ is a function of S, hence random, and the quantity to compute is the expected risk

$$\mathbb{E}_S\big[L_D(h_{\hat{w}})\big] = \mathbb{E}_S\Big[\mathbb{E}_{(x,y)\sim\mathcal{D}}\big[(y - x^\top \hat{w})^2\big]\Big],$$

with the test pair $(x, y) \sim \mathcal{D}$ independent of S.

2.2 Bias-variance decomposition

Notation. For a positive semidefinite Σ let $\|a\|_\Sigma^2 = a^\top \Sigma a$, and let $\mathrm{trace}(M) = \sum_i M_{ii}$. The covariance matrix of the estimator is

$$\mathrm{Cov}_S(\hat{w}) = \mathbb{E}_S\left[(\hat{w} - \mathbb{E}_S[\hat{w}])(\hat{w} - \mathbb{E}_S[\hat{w}])^\top\right].$$

Theorem 2.1 (Risk of a learned linear model)**.** *Under the data model above, let $\hat{w}$ be learned from $S \sim \mathcal{D}^m$ by any learner. Then*

$$\begin{aligned}\mathbb{E}_S\left[L_{\mathcal{D}}(h_{\hat{w}})\right] &= \mathbb{E}_S\left[\|\hat{w} - w^*\|_\Sigma^2\right] + \sigma^2 \\ &= \underbrace{\|\mathbb{E}_S[\hat{w}] - w^*\|_\Sigma^2}_{\textit{bias}} + \underbrace{\mathbb{E}_S\left[\|\hat{w} - \mathbb{E}_S[\hat{w}]\|_\Sigma^2\right]}_{\textit{variance}} + \underbrace{\sigma^2}_{\textit{Bayes risk}} \\ &= \|\mathbb{E}_S[\hat{w}] - w^*\|_\Sigma^2 + \mathrm{trace}\left(\mathrm{Cov}_S(\hat{w})\,\Sigma\right) + \sigma^2.\end{aligned}$$

Proof. First equality. Fix S, so that $\hat{w}$ is a constant for the independent test pair. Inserting the model and expanding the square, with $(x^\top a)^2 = a^\top x x^\top a$,

$$\begin{aligned}L_{\mathcal{D}}(h_{\hat{w}}) &= \mathbb{E}_{x,\epsilon}\left[(x^\top \hat{w} - x^\top w^* - \epsilon)^2\right] = \mathbb{E}_{x,\epsilon}\left[(x^\top(\hat{w} - w^*) - \epsilon)^2\right] \\ &= \mathbb{E}_x\left[(\hat{w} - w^*)^\top x x^\top (\hat{w} - w^*)\right] - 2\,\mathbb{E}_{x,\epsilon}\left[\epsilon\, x^\top(\hat{w} - w^*)\right] + \mathbb{E}[\epsilon^2] && \text{(linearity)} \\ &= (\hat{w} - w^*)^\top \mathbb{E}_x[x x^\top](\hat{w} - w^*) - 2\,\mathbb{E}[\epsilon]\,\mathbb{E}_x\left[x^\top(\hat{w} - w^*)\right] + \sigma^2 && (x,\ \epsilon \text{ independent}) \\ &= \|\hat{w} - w^*\|_\Sigma^2 + \sigma^2 && (\mathbb{E}[\epsilon] = 0).\end{aligned}$$

Taking $\mathbb{E}_S$ on both sides gives the claim.

Second equality. Split $\hat{w} - w^* = u + v$ with $u = \hat{w} - \mathbb{E}_S[\hat{w}]$ random and $v = \mathbb{E}_S[\hat{w}] - w^*$ deterministic. Since Σ is symmetric,

$$\|u + v\|_\Sigma^2 = u^\top \Sigma u + v^\top \Sigma v + 2\, u^\top \Sigma v.$$

By linearity $\mathbb{E}_S[u^\top \Sigma v] = \mathbb{E}_S[u]^\top \Sigma v = 0$, because $\mathbb{E}_S[u] = \mathbb{E}_S[\hat{w}] - \mathbb{E}_S[\hat{w}] = 0$. Hence $\mathbb{E}_S\|\hat{w} - w^*\|_\Sigma^2 = \mathbb{E}_S\|u\|_\Sigma^2 + \|v\|_\Sigma^2$, which is variance plus bias.

Third equality. A scalar equals its trace, and the trace is cyclic and commutes with expectation (both are linear):

$$\mathbb{E}_S\left[u^\top \Sigma u\right] = \mathbb{E}_S\left[\,\mathrm{trace}(u^\top \Sigma u)\right] = \mathbb{E}_S\left[\,\mathrm{trace}(u u^\top \Sigma)\right] = \mathrm{trace}\left(\mathbb{E}_S[u u^\top]\,\Sigma\right) = \mathrm{trace}\left(\mathrm{Cov}_S(\hat{w})\,\Sigma\right). \quad \square$$

Bias and variance are ≥ 0 (Σ is positive semidefinite), so no learned linear model beats the Bayes risk σ^2, which h_{w^*} attains. An **unbiased** estimator, $\mathbb{E}_S[\hat{w}] = w^*$, removes the bias term; still, a biased estimator with small enough $\mathrm{Cov}_S(\hat{w})$ can have a smaller expected risk. Balancing the two terms is the **bias-variance tradeoff**.

2.3 Risk of OLS and consistency

From now on $\hat{w}$ is the OLS estimator.

Theorem 2.2 (OLS is unbiased)**.** *If $\hat{\Sigma}$ is invertible with probability* 1*, then $\mathbb{E}_S[\hat{w}] = w^*$.*

Proof. Given $x_1, \ldots, x_m$, the matrix $\hat{\Sigma}$ is fixed, and $\mathbb{E}[y_i \mid x_1, \ldots, x_m] = x_i^\top w^* + \mathbb{E}[\epsilon_i] = x_i^\top w^*$ because ϵ_i is independent of the inputs and has mean 0. Therefore

$$\begin{aligned}\mathbb{E}_S[\hat{w}] &= \mathbb{E}_{x_1,\ldots,x_m}\left[\mathbb{E}\left[\hat{w} \mid x_1, \ldots, x_m\right]\right] && \text{(tower property)} \\ &= \mathbb{E}_{x_1,\ldots,x_m}\left[\hat{\Sigma}^{-1}\left(\frac{1}{m}\sum_{i=1}^m x_i\,\mathbb{E}[y_i \mid x_1, \ldots, x_m]\right)\right] && \text{(linearity)} \\ &= \mathbb{E}_{x_1,\ldots,x_m}\left[\hat{\Sigma}^{-1}\left(\frac{1}{m}\sum_{i=1}^m x_i x_i^\top\right) w^*\right] = \mathbb{E}_{x_1,\ldots,x_m}\left[\hat{\Sigma}^{-1}\hat{\Sigma}\, w^*\right] = w^*. && \square\end{aligned}$$

Theorem 2.3 (Covariance of the OLS estimator). *If $\hat{\Sigma}$ is invertible with probability 1, then*

$$\mathrm{Cov}_S(\hat{w}) = \mathbb{E}_S\left[(\hat{w} - w^*)(\hat{w} - w^*)^\top\right] = \frac{\sigma^2}{m}\,\mathbb{E}_{x_1,\dots,x_m}\left[\hat{\Sigma}^{-1}\right].$$

Proof. The first equality is the definition of Cov_S with $\mathbb{E}_S[\hat{w}] = w^*$ (theorem 2.2). Substituting $y_i = x_i^\top w^* + \epsilon_i$,

$$\frac{1}{m}\sum_{i=1}^m y_i x_i = \Big(\frac{1}{m}\sum_{i=1}^m x_i x_i^\top\Big) w^* + \frac{1}{m}\sum_{i=1}^m x_i \epsilon_i = \hat{\Sigma} w^* + \frac{1}{m}\sum_{i=1}^m x_i \epsilon_i \quad\Longrightarrow\quad \hat{w} - w^* = \hat{\Sigma}^{-1}\Big(\frac{1}{m}\sum_{i=1}^m x_i \epsilon_i\Big).$$

Since $\hat{\Sigma}^{-1}$ is symmetric, $(\hat{w} - w^*)^\top = \big(\frac{1}{m}\sum_j x_j^\top \epsilon_j\big)\hat{\Sigma}^{-1}$, so

$$(\hat{w} - w^*)(\hat{w} - w^*)^\top = \frac{1}{m^2}\,\hat{\Sigma}^{-1}\Big(\sum_{i=1}^m\sum_{j=1}^m \epsilon_i \epsilon_j\, x_i x_j^\top\Big)\hat{\Sigma}^{-1}.$$

The ϵ_i are i.i.d. and independent of the inputs, so $\mathbb{E}[\epsilon_i \epsilon_j \mid x_1, \dots, x_m] = \mathbb{E}[\epsilon_i \epsilon_j]$, which is $\mathbb{E}[\epsilon_i]\mathbb{E}[\epsilon_j] = 0$ for $i \neq j$ and $\mathbb{E}[\epsilon_i^2] = \sigma^2$ for $i = j$. Conditioning on the inputs (tower property) and using linearity,

$$\begin{aligned}
\mathbb{E}_S\left[(\hat{w} - w^*)(\hat{w} - w^*)^\top\right] &= \frac{1}{m^2}\,\mathbb{E}_{x_1,\dots,x_m}\Big[\hat{\Sigma}^{-1}\Big(\sum_{i=1}^m\sum_{j=1}^m \mathbb{E}[\epsilon_i\epsilon_j]\, x_i x_j^\top\Big)\hat{\Sigma}^{-1}\Big] \\
&= \frac{1}{m^2}\,\mathbb{E}_{x_1,\dots,x_m}\Big[\hat{\Sigma}^{-1}\Big(\sigma^2 \sum_{i=1}^m x_i x_i^\top\Big)\hat{\Sigma}^{-1}\Big] \qquad \text{(only } i = j \text{ survives)} \\
&= \frac{1}{m^2}\,\mathbb{E}_{x_1,\dots,x_m}\left[\hat{\Sigma}^{-1}\big(\sigma^2 m \hat{\Sigma}\big)\hat{\Sigma}^{-1}\right] = \frac{\sigma^2}{m}\,\mathbb{E}_{x_1,\dots,x_m}\left[\hat{\Sigma}^{-1}\right].
\end{aligned}$$

□

Gaussian features. Let $\mathcal{D}_\mathcal{X} = \mathcal{N}(0, \Sigma)$ with $\Sigma \in \mathbb{R}^{p\times p}$ invertible (the mean is 0, so $\Sigma = \mathbb{E}[xx^\top]$ as in the data model), and $y = x^\top w^* + \epsilon$ as before. We use without proof the following facts on the Wishart distribution, for $x_1, \dots, x_m$ i.i.d. $\mathcal{N}(0, \Sigma)$. If $m \geq p$, then $\hat{\Sigma}$ is invertible with probability 1; $\hat{\Sigma}$ is Wishart with m degrees of freedom and scale matrix $\frac{1}{m}\Sigma$; $\hat{\Sigma}^{-1}$ is inverse Wishart with m degrees of freedom and scale matrix $m\Sigma^{-1}$; and for $m > p + 1$,

$$\mathbb{E}\left[\hat{\Sigma}^{-1}\right] = \frac{m}{m - p - 1}\,\Sigma^{-1}.$$

Theorem 2.4 (Risk of OLS for Gaussian features and additive noise). *For $\mathcal{D}$ as above and $m > p + 1$,*

$$\mathbb{E}_S\left[L_\mathcal{D}(h_{\hat{w}})\right] = \underbrace{\frac{\sigma^2 p}{m - p - 1}}_{\text{variance}} + \underbrace{\sigma^2}_{\text{Bayes risk}} = \frac{\sigma^2(m-1)}{m - p - 1}.$$

In particular, for fixed p, $\mathbb{E}_S[L_\mathcal{D}(h_{\hat{w}})] \to \sigma^2 = L_\mathcal{D}^*$ as $m \to \infty$.

Proof. Since $m > p + 1 \geq p$, $\hat{\Sigma}$ is invertible with probability 1, so theorems 2.2 and 2.3 apply. Starting from theorem 2.1,

$$\begin{aligned}
\mathbb{E}_S\left[L_\mathcal{D}(h_{\hat{w}})\right] &= \|\mathbb{E}_S[\hat{w}] - w^*\|_\Sigma^2 + \mathrm{trace}\left(\mathrm{Cov}_S(\hat{w})\,\Sigma\right) + \sigma^2 \\
&= 0 + \mathrm{trace}\left(\frac{\sigma^2}{m}\,\mathbb{E}\left[\hat{\Sigma}^{-1}\right]\Sigma\right) + \sigma^2 && \text{(theorems 2.2 and 2.3)} \\
&= \mathrm{trace}\left(\frac{\sigma^2}{m}\cdot\frac{m}{m - p - 1}\,\Sigma^{-1}\Sigma\right) + \sigma^2 && \text{(Wishart mean)} \\
&= \frac{\sigma^2}{m - p - 1}\,\mathrm{trace}(I_p) + \sigma^2 = \frac{\sigma^2 p}{m - p - 1} + \sigma^2 && (\mathrm{trace}(I_p) = p) \\
&= \frac{\sigma^2(p + m - p - 1)}{m - p - 1} = \frac{\sigma^2(m-1)}{m - p - 1}.
\end{aligned}$$

For fixed p, $\frac{m-1}{m-p-1} \to 1$ as $m \to \infty$. □

Consistency. Let $\mathcal{D}$ be a distribution on $\mathcal{X} \times \mathcal{Y}$, ℓ a loss with Bayes risk $L^*_{\mathcal{D}}$, and $\hat{h}_S$ the predictor that a learner $\mathcal{A}$ returns on $S \sim \mathcal{D}^m$.

Definition 2.5 (Consistency)**.** $\mathcal{A}$ is **consistent** for $\mathcal{D}$ and ℓ if $\mathbb{E}_{S\sim\mathcal{D}^m}\left[L_{\mathcal{D}}(\hat{h}_S)\right] \to L^*_{\mathcal{D}}$ as $m \to \infty$. It is **universally consistent** if it is consistent for every distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$.

By theorem 2.4, OLS is consistent for $x \sim \mathcal{N}(0, \Sigma)$, $y = x^\top w^* + \epsilon$ (with p fixed). It is not universally consistent: it fails as soon as the true model is not linear, for instance $y = (x^\top w^*)^2 + \epsilon$.

Remark. Take $x \sim \mathcal{N}(0, \Sigma)$ and $w^* \neq 0$ in this example. Then $\mathbb{E}[y \mid x] = (x^\top w^*)^2$ and $\mathrm{Var}(y \mid x) = \sigma^2$, so $L^*_{\mathcal{D}} = \sigma^2$. For any w put $Z = x^\top w - (x^\top w^*)^2$; the decomposition in the proof of proposition 1.8, $\mathbb{E}[Z^2] \geq (\mathbb{E}Z)^2$, $\mathbb{E}[x] = 0$ and $\mathbb{E}[(x^\top w^*)^2] = w^{*\top}\mathbb{E}[xx^\top]w^*$ give

$$L_{\mathcal{D}}(h_w) = \mathbb{E}[Z^2] + \sigma^2 \geq (\mathbb{E}Z)^2 + \sigma^2 = \left(0 - \|w^*\|^2_\Sigma\right)^2 + \sigma^2 = \|w^*\|^4_\Sigma + \sigma^2.$$

Hence $\mathbb{E}_S[L_{\mathcal{D}}(h_{\hat{w}})] \geq L^*_{\mathcal{D}} + \|w^*\|^4_\Sigma$ for every m, and $\|w^*\|_\Sigma > 0$ since Σ is invertible.

3 Nearest-neighbour classification and consistency

3.1 Nearest-neighbour rules and their asymptotic risk

The rules. Let $\mathcal{X} \subseteq \mathbb{R}^p$ with the Euclidean distance and $S = ((x_i, y_i))_{i=1}^m$. For a test point x, sort the sample points by distance to x and let $\pi_k(x) \in \{1, \dots, m\}$ be the index of the k**-th nearest neighbour** of x, so that

$$\|x - x_{\pi_1(x)}\| \leq \|x - x_{\pi_2(x)}\| \leq \cdots \leq \|x - x_{\pi_m(x)}\|.$$

The **nearest-neighbour (NN) rule** predicts the label of the closest point, $h^{\mathrm{NN}}_S(x) = y_{\pi_1(x)}$; the k**-NN rule** $h^{k\mathrm{NN}}_S(x)$ predicts the majority vote of $y_{\pi_1(x)}, \dots, y_{\pi_k(x)}$, which for labels ± 1 reads $h^{k\mathrm{NN}}_S(x) = \mathrm{sign}\left(\frac{1}{k}\sum_{i=1}^k y_{\pi_i(x)}\right)$. Ties (in distances or in the vote) are not treated. The 1-NN rule is constant on the **Voronoi cells** of the sample (the sets of points closer to x_i than to any other x_j). For finite m the decision boundary depends strongly on S; for large m these rules can represent very complex boundaries (fig. 2).

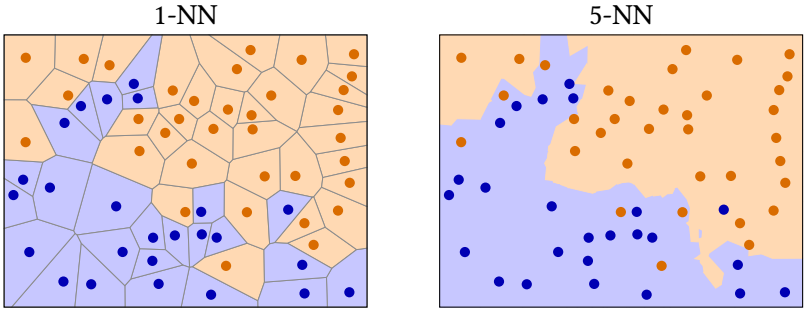


Figure 2: 1-NN and 5-NN decision regions on the same sample.

Example 3.1 (Installation crash, simulated)**.** Take $\mathcal{D}_{\mathcal{X}} = \mathrm{Uniform}[0, 1]$ and $\eta(x) = |1 - 2x|$, whose Bayes risk is $L^*_{\mathcal{D}} = \frac{1}{4}$ (example 1.7). Figure 3 shows the test error of k-NN (5000 test points, average over 25 samples) as m grows. With $k = 1$ or $k = 3$ fixed, the error levels off strictly above $L^*_{\mathcal{D}}$; with $k \approx (\log m)^2$ it approaches $L^*_{\mathcal{D}}$; with $k \approx m/2$ it stays high. The results below explain all three behaviours.

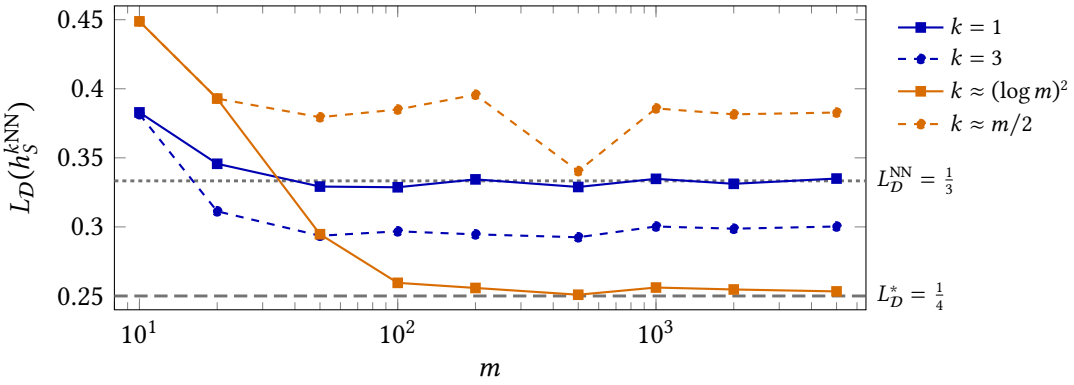


Figure 3: Test error of k-NN on the installation-crash example.

Asymptotic risk. Recall from theorem 1.6 that $L_D^* = \mathbb{E}_x[\min\{\eta(x), 1-\eta(x)\}]$, attained by $h^*(x) = \mathbb{1}\{\eta(x) \geq \frac{1}{2}\}$, which requires knowing η.

Theorem 3.2 (Asymptotic expected risk of the NN rule). *Let* $L_D^{NN} = \mathbb{E}_{x\sim D_X}\big[2\eta(x)(1-\eta(x))\big]$. *Then*

$$\mathbb{E}_{S\sim D^m}\big[L_D(h_S^{NN})\big] \longrightarrow L_D^{NN} \quad \text{as } m \to \infty, \qquad \text{and} \qquad L_D^* \leq L_D^{NN} \leq 2L_D^*.$$

The convergence is proved later in this subsection; the inequalities follow directly.

Proof of the inequalities. Fix x and let $a = \min\{\eta(x), 1-\eta(x)\} \leq \frac{1}{2}$. Since $\{\eta(x), 1-\eta(x)\} = \{a, 1-a\}$, we have $2\eta(x)(1-\eta(x)) = 2a(1-a)$, and $\frac{1}{2} \leq 1-a \leq 1$ gives

$$a = 2a \cdot \tfrac{1}{2} \leq 2a(1-a) \leq 2a.$$

Taking $\mathbb{E}_x$ gives $L_D^* \leq L_D^{NN} \leq 2L_D^*$. □

So with infinitely many data the NN rule is at most twice as bad as the Bayes classifier.

Example 3.3 (Installation crash, NN limit). In example 3.1, $u = |1-2x|$ is again uniform on $[0,1]$: for $t \in [0,1]$, $\mathbb{P}(u \leq t) = \mathbb{P}\big(\frac{1-t}{2} \leq x \leq \frac{1+t}{2}\big) = t$. Hence, with $\mathbb{E}[u] = \frac{1}{2}$ and $\mathbb{E}[u^2] = \frac{1}{3}$,

$$L_D^{NN} = \mathbb{E}\big[2u(1-u)\big] = 2\Big(\frac{1}{2} - \frac{1}{3}\Big) = \frac{1}{3},$$

the level reached by the $k = 1$ curve in fig. 3, and indeed $\frac{1}{4} \leq \frac{1}{3} \leq \frac{1}{2}$.

Theorem 3.4 (Asymptotic expected risk of the k-NN rule). *Let k be fixed. The limit below exists and satisfies*

$$L_D^{kNN} = \lim_{m\to\infty} \mathbb{E}_{S\sim D^m}\big[L_D(h_S^{kNN})\big], \qquad L_D^* \leq L_D^{kNN} \leq \Big(1 + \frac{2}{\sqrt{k}}\Big)L_D^*.$$

The proof (omitted) follows the case $k = 1$ but is more involved. The upper bound approaches L_D^* only as $k \to \infty$, which suggests letting k grow with m.

Why nearest neighbours work. For large m the nearest neighbours of x lie close to x, so (if η is regular) their labels have nearly the same law as the label of x. To make this precise, let $D_X(x;\epsilon) = \mathbb{P}_{x'\sim D_X}(\|x'-x\| \leq \epsilon)$ be the mass of the closed ϵ-ball around x, and let $\mathrm{support}(D_X)$ be the set of $x \in \mathcal{X}$ with $D_X(x;\epsilon) > 0$ for every $\epsilon > 0$. Random vectors z_m **converge in probability** to z if $\lim_{m\to\infty} \mathbb{P}(\|z_m - z\| > \epsilon) = 0$ for every $\epsilon > 0$.

Lemma 3.5 (Nearest neighbours converge in probability). *Let $x \in \mathrm{support}(D_X)$, $x_1, \dots, x_m$ i.i.d. D_X, and $k/m \to 0$ as $m \to \infty$. Then $\mathbb{P}(\|x_{\pi_k(x)} - x\| > \epsilon) \to 0$ for every $\epsilon > 0$. Since $\|x_{\pi_j(x)} - x\| \leq \|x_{\pi_k(x)} - x\|$ for $j \leq k$, all of $x_{\pi_1(x)}, \dots, x_{\pi_k(x)}$ converge to x in probability.*

Proof. Fix $\epsilon > 0$, let $p = \mathcal{D}_{\mathcal{X}}(x;\epsilon) > 0$ and let $Z = \sum_{i=1}^{m} \mathbb{1}\{\|x_i - x\| \le \epsilon\}$ count the sample points in the ϵ-ball. The indicators are independent Bernoulli(p), so $Z \sim \text{Binomial}(m,p)$ and $\mathbb{E}[Z] = mp$. The k-th neighbour is farther than ϵ exactly when fewer than k points lie in the ball:

$$\mathbb{P}\big(\|x_{\pi_k(x)} - x\| > \epsilon\big) = \mathbb{P}(Z \le k-1).$$

For $k = 1$ this is explicit: by independence $\mathbb{P}(Z=0) = \prod_{i=1}^{m} \mathbb{P}(\|x_i - x\| > \epsilon) = (1-p)^m \to 0$. In general we use **Hoeffding's inequality**: for a sum Z of m independent variables with values in $[0,1]$, $\mathbb{P}(Z \le \mathbb{E}[Z] - t) \le e^{-2t^2/m}$ for $t > 0$. Since $k/m \to 0$, for m large enough $\frac{k-1}{m} < \frac{p}{2}$, i.e. $k - 1 < mp - \frac{mp}{2}$, hence

$$\mathbb{P}(Z \le k-1) \le \mathbb{P}\Big(Z \le mp - \frac{mp}{2}\Big) \le \exp\Big(-\frac{2}{m}\Big(\frac{mp}{2}\Big)^2\Big) = e^{-mp^2/2} \xrightarrow[m\to\infty]{} 0. \qquad \square$$

Proof of the convergence in theorem 3.2. We add two assumptions, which are not necessary but simplify the proof:

(A1) $\eta : \mathcal{X} \to [0,1]$ is **uniformly continuous**: for every $\delta > 0$ there is $\epsilon_\delta > 0$ such that $\|x - x'\| \le \epsilon_\delta$ implies $|\eta(x) - \eta(x')| \le \delta$; equivalently, $|\eta(x) - \eta(x')| > \delta$ implies $\|x - x'\| > \epsilon_\delta$.

(A2) $\text{support}(\mathcal{D}_{\mathcal{X}}) = \mathcal{X}$.

Step 1: conditional risk. Under the 0-1 loss, $L_D(h_S^{\text{NN}}) = \mathbb{E}_{(x,y)\sim D}\big[\mathbb{1}\{y \ne y_{\pi_1(x)}\}\big]$. Generate the $m+1$ pairs $(x,y), (x_1,y_1), \dots, (x_m,y_m)$ by first drawing the inputs $x, x_1, \dots, x_m$ i.i.d. from $\mathcal{D}_{\mathcal{X}}$ and then each label independently, equal to 1 with probability η of its input. By the tower property,

$$\mathbb{E}_S\big[L_D(h_S^{\text{NN}})\big] = \mathbb{E}_{S,(x,y)\sim D^{m+1}}\big[\mathbb{1}\{y \ne y_{\pi_1(x)}\}\big] = \mathbb{E}_{x,x_1,\dots,x_m}\Big[\mathbb{E}\big[\mathbb{1}\{y \ne y_{\pi_1(x)}\} \,\big|\, x, x_1, \dots, x_m\big]\Big].$$

Given the inputs, $\pi_1(x)$ is fixed and the labels y, $y_{\pi_1(x)}$ are independent with $\mathbb{P}(y=1) = \eta(x)$, $\mathbb{P}(y_{\pi_1(x)} = 1) = \eta(x_{\pi_1(x)})$. They differ if they are $(1,0)$ or $(0,1)$, so, writing $\eta = \eta(x)$ and $\eta' = \eta(x_{\pi_1(x)})$,

$$\mathbb{E}\big[\mathbb{1}\{y \ne y_{\pi_1(x)}\} \,\big|\, x, x_1, \dots, x_m\big] = \eta(1-\eta') + (1-\eta)\eta'.$$

Step 2: reduction to $|\eta' - \eta|$. Subtracting the integrand of L_D^{NN},

$$\begin{aligned}\eta(1-\eta') + (1-\eta)\eta' - 2\eta(1-\eta) &= \eta - \eta\eta' + \eta' - \eta\eta' - 2\eta + 2\eta^2 \\ &= (\eta' - \eta) - 2\eta(\eta' - \eta) = (1-2\eta)(\eta' - \eta).\end{aligned}$$

Using $|\mathbb{E}z| \le \mathbb{E}|z|$ (Jensen's inequality, $|\cdot|$ convex) and $|1-2\eta| \le 1$ (as $\eta \in [0,1]$),

$$\Big|\mathbb{E}_S\big[L_D(h_S^{\text{NN}})\big] - L_D^{\text{NN}}\Big| = \Big|\mathbb{E}\big[(1-2\eta)(\eta' - \eta)\big]\Big| \le \mathbb{E}\big[|1-2\eta|\,|\eta' - \eta|\big] \le \mathbb{E}_{x,x_1,\dots,x_m}|\eta(x_{\pi_1(x)}) - \eta(x)|.$$

Step 3: splitting at δ. Fix $\delta \in (0,1)$. Split according to whether $|\eta' - \eta| > \delta$; on that event bound $|\eta' - \eta| \le 1$, on its complement by δ:

$$\mathbb{E}|\eta' - \eta| = \mathbb{E}\big[|\eta' - \eta|\,\mathbb{1}\{|\eta' - \eta| > \delta\}\big] + \mathbb{E}\big[|\eta' - \eta|\,\mathbb{1}\{|\eta' - \eta| \le \delta\}\big] \le \mathbb{P}\big(|\eta' - \eta| > \delta\big) + \delta.$$

Step 4: the bad event vanishes. By (A1), $|\eta' - \eta| > \delta$ forces $\|x_{\pi_1(x)} - x\| > \epsilon_\delta$, and the nearest neighbour is farther than ϵ_δ iff all x_i are; conditioning on x and using the case $k=1$ of the proof of lemma 3.5,

$$\mathbb{P}\big(|\eta' - \eta| > \delta\big) \le \mathbb{P}\big(\|x_{\pi_1(x)} - x\| > \epsilon_\delta\big) = \mathbb{E}_x\Big[\mathbb{P}\Big(\min_{i\le m}\|x_i - x\| > \epsilon_\delta \,\Big|\, x\Big)\Big] = \mathbb{E}_x\Big[\big(1 - \mathcal{D}_{\mathcal{X}}(x;\epsilon_\delta)\big)^m\Big].$$

By (A2) every x has $\mathcal{D}_{\mathcal{X}}(x;\epsilon_\delta) > 0$, so the integrand tends to 0 for every x; it lies in $[0,1]$, hence its expectation tends to 0 as well.

Conclusion. Combining Steps 2–4, $\limsup_{m\to\infty} \big|\mathbb{E}_S[L_D(h_S^{\text{NN}})] - L_D^{\text{NN}}\big| \le \delta$ for every $\delta \in (0,1)$, so the limit is 0. $\square$

Without assumptions. The general proof follows Steps 1–2 and then shows $\mathbb{E}|\eta(x_{\pi_1(x)}) - \eta(x)| \to 0$ via part (ii) of the following lemma with $k = 1$ and $f = \eta$, which needs no assumption on η or D_X; theorem 3.4 is proved along the same route. The proof of the lemma is omitted (Devroye, Györfi, Lugosi, Lemmas 5.3–5.4).

Lemma 3.6 (Convergence of a function evaluated at the k nearest neighbours). *Let $f : \mathcal{X} \to \mathbb{R}$ with $\mathbb{E}_{x\sim D_X}|f(x)| < \infty$, and $x, x_1, \dots, x_m$ i.i.d. D_X.*

(i) ***(Stone's lemma)*** $\frac{1}{k}\sum_{i=1}^{k} \mathbb{E}|f(x_{\pi_i(x)})| \le \gamma_p\, \mathbb{E}|f(x)|$, *with* $\gamma_p = \left(1 + \frac{2}{\sqrt{2-\sqrt{3}}}\right)^p - 1$.

(ii) If $k/m \to 0$, then $\mathbb{E}_{x,x_1,\dots,x_m}\left[\frac{1}{k}\sum_{i=1}^{k}|f(x_{\pi_i(x)}) - f(x)|\right] \to 0$ *as* $m \to \infty$.

3.2 Plug-in classifiers and universal consistency

We use consistency and universal consistency as in definition 2.5, for the 0-1 loss.

Plug-in classifiers. The Bayes classifier $h^*(x) = \mathbb{1}\{\eta(x) \ge \frac{1}{2}\}$ (or $\operatorname{sign}(\eta(x) - \frac{1}{2})$ for labels ± 1) is out of reach because η is unknown. A **plug-in classifier** estimates η by some $\hat\eta$ computed from S and predicts

$$\hat h(x) = \mathbb{1}\{\hat\eta(x) \ge \tfrac{1}{2}\} \qquad \big(\text{or } \operatorname{sign}(\hat\eta(x) - \tfrac{1}{2})\big).$$

The idea behind its consistency: if $\hat\eta(x) \to \eta(x)$ for (almost) all x as $m \to \infty$, then $L_D(\hat h) \to L_D^*$; lemma 3.10 makes this quantitative.

Example 3.7 (Naive Bayes). By Bayes' theorem, with $\mathbb{P}(x \mid y = i)$ the class-conditional probability or density,

$$\eta(x) = \frac{\mathbb{P}(y=1)\mathbb{P}(x \mid y=1)}{\mathbb{P}(x)}, \qquad 1 - \eta(x) = \frac{\mathbb{P}(y=0)\mathbb{P}(x \mid y=0)}{\mathbb{P}(x)}.$$

Since $\eta(x) \ge \frac{1}{2} \iff \eta(x) \ge 1 - \eta(x)$, multiplying by $\mathbb{P}(x) > 0$ gives $h^*(x) = \mathbb{1}\{\mathbb{P}(y=1)\mathbb{P}(x \mid y=1) \ge \mathbb{P}(y=0)\mathbb{P}(x \mid y=0)\}$. The plug-in version is $\hat h(x) = \mathbb{1}\{\hat p_1 \hat b_1(x) \ge \hat p_0 \hat b_0(x)\}$, where $\hat p_i$ is the fraction of training points with label i (an easy estimate of $\mathbb{P}(y = i)$) and $\hat b_i(x)$ estimates $\mathbb{P}(x \mid y = i)$. The latter is hard to estimate when x is high-dimensional; **Naive Bayes** assumes that the coordinates of $x = (x^{(1)}, \dots, x^{(p)})$ are independent given the label,

$$\mathbb{P}(x \mid y = i) = \prod_{j=1}^{p} \mathbb{P}\big(x^{(j)} \mid y = i\big),$$

so that $\hat b_i(x)$ only needs one-dimensional estimates.

Weighted-average estimates. For a test point x, let $w_1(x), \dots, w_m(x) \in [0,1]$ be weights with $\sum_{i=1}^m w_i(x) = 1$, computed from x and $x_1, \dots, x_m$ (not from the labels), and set

$$\hat\eta(x) = \sum_{i=1}^{m} \mathbb{1}\{y_i = 1\}\, w_i(x), \qquad \hat h(x) = \mathbb{1}\{\hat\eta(x) \ge \tfrac{1}{2}\}.$$

The NN rule is the case $w_i(x) = \mathbb{1}\{i = \pi_1(x)\}$: then $\hat\eta(x) = \mathbb{1}\{y_{\pi_1(x)} = 1\}$ and $\hat h(x) = y_{\pi_1(x)}$. The k-NN rule is the case $w_i(x) = \frac{1}{k}$ if x_i is one of the k nearest neighbours of x and 0 otherwise:

$$\hat\eta(x) = \frac{1}{k}\sum_{j=1}^{k} \mathbb{1}\{y_{\pi_j(x)} = 1\} = \sum_{i=1}^{m} \mathbb{1}\{y_i = 1\}\, \frac{\mathbb{1}\{i \in \{\pi_1(x), \dots, \pi_k(x)\}\}}{k},$$

and $\hat h(x) = \mathbb{1}\{\hat\eta(x) \ge \frac{1}{2}\}$ is the majority vote.

Theorem 3.8 (Stone's consistency theorem). *The plug-in classifier $\hat{h}$ built from weights w_i as above is universally consistent if, with $x, x_1, \dots, x_m$ i.i.d. $\mathcal{D}_{\mathcal{X}}$:*

(i) there is c such that, for every $f \geq 0$ with $\mathbb{E}_{x\sim D_{\mathcal{X}}}[f(x)] < \infty$, $\mathbb{E}\Big[\sum_{i=1}^{m} w_i(x)\, f(x_i)\Big] \leq c\, \mathbb{E}_{x\sim D_{\mathcal{X}}}[f(x)]$;

(ii) for every $a > 0$, $\lim_{m\to\infty} \mathbb{E}\Big[\sum_{i=1}^{m} w_i(x)\, \mathbb{1}\{\|x_i - x\| > a\}\Big] = 0$;

(iii) $\lim_{m\to\infty} \mathbb{E}\Big[\max_{i\leq m} w_i(x)\Big] = 0$.

The proof is in section 3.3: condition (ii) makes the weights concentrate near x, (iii) forbids any single label from dominating, and (i) is needed only when η is not uniformly continuous.

Universal consistency of k-NN.

Theorem 3.9 (Universal consistency of k-NN). *If $k \to \infty$ and $k/m \to 0$ as $m \to \infty$, then for every distribution $\mathcal{D}$,*

$$\mathbb{E}_{S\sim D^m}\left[L_D(h_S^{k\mathrm{NN}})\right] \to L_D^* \qquad \text{as } m \to \infty.$$

Proof. We check the conditions of theorem 3.8 for the k-NN weights $w_i(x) = \frac{1}{k}\mathbb{1}\{i \in \{\pi_1(x), \dots, \pi_k(x)\}\}$.

(i) The sum only picks the k nearest neighbours, $\sum_{i=1}^{m} w_i(x) f(x_i) = \frac{1}{k}\sum_{j=1}^{k} f(x_{\pi_j(x)})$. Taking expectations, Stone's lemma (lemma 3.6(i), with $|f| = f$) gives the bound with $c = \gamma_p$.

(ii) If the k-th neighbour is within distance a, so are the closer ones; hence the fraction of the k neighbours farther than a is 0 unless $\|x_{\pi_k(x)} - x\| > a$, and it is always ≤ 1:

$$\sum_{i=1}^{m} w_i(x)\mathbb{1}\{\|x_i - x\| > a\} = \frac{1}{k}\sum_{j=1}^{k} \mathbb{1}\{\|x_{\pi_j(x)} - x\| > a\} \leq \mathbb{1}\{\|x_{\pi_k(x)} - x\| > a\}.$$

So the expectation is at most $\mathbb{E}_x\left[\mathbb{P}(\|x_{\pi_k(x)} - x\| > a \mid x)\right]$. Since $k/m \to 0$, the inner probability tends to 0 for every $x \in \mathrm{support}(\mathcal{D}_{\mathcal{X}})$ (lemma 3.5), which carries all the mass of $\mathcal{D}_{\mathcal{X}}$; it is bounded by 1, so the expectation tends to 0.

(iii) $\max_i w_i(x) = \frac{1}{k} \to 0$ because $k \to \infty$. □

This explains fig. 3: $k \approx (\log m)^2$ satisfies both conditions and reaches L_D^*; fixed k violates $k \to \infty$ and stays at $L_D^{k\mathrm{NN}}$ (for $k = 1$, at $L_D^{\mathrm{NN}} = \frac{1}{3}$); $k \approx m/2$ violates $k/m \to 0$.

3.3 Proof of Stone's theorem (not examinable)

This subsection is not part of the exam.

Lemma 3.10 (Risk bound for plug-in classifiers). *Under the 0-1 loss, let $\hat{h}(x) = \mathbb{1}\{\hat{\eta}(x) \geq \frac{1}{2}\}$ for a fixed function $\hat{\eta} : \mathcal{X} \to [0,1]$. Then*

$$L_D(\hat{h}) - L_D^* \leq 2\,\mathbb{E}_{x\sim D_{\mathcal{X}}}|\hat{\eta}(x) - \eta(x)| \leq 2\sqrt{\mathbb{E}_{x\sim D_{\mathcal{X}}}\left[(\hat{\eta}(x) - \eta(x))^2\right]}.$$

Proof. By theorem 1.5, at each x a classifier pays $\eta(x)$ for predicting 0 and $1 - \eta(x)$ for predicting 1. Where $\hat{h}(x) = h^*(x)$ the two classifiers pay the same; where they differ, h^* pays the smaller of the two amounts and $\hat{h}$ the larger, which exceeds it by $|(1 - \eta(x)) - \eta(x)| = 2|\eta(x) - \frac{1}{2}|$. Hence

$$L_D(\hat{h}) - L_D^* = 2\,\mathbb{E}_x\Big[\big|\eta(x) - \tfrac{1}{2}\big|\,\mathbb{1}\{h^*(x) \neq \hat{h}(x)\}\Big].$$

If $h^*(x) \neq \hat{h}(x)$, one of $\eta(x)$, $\hat{\eta}(x)$ is $\geq \frac{1}{2}$ and the other $< \frac{1}{2}$, so $\frac{1}{2}$ lies between them and $|\eta(x) - \frac{1}{2}| \leq |\eta(x) - \hat{\eta}(x)|$. Bounding the indicator by 1 gives the first inequality. The second is Jensen's inequality for the convex $z \mapsto z^2$, $(\mathbb{E}|z|)^2 \leq \mathbb{E}[z^2]$, followed by a square root. □

Reduction. Apply lemma 3.10 for each sample S (with $\hat{\eta}$ computed from S), take $\mathbb{E}_S$, and use Jensen once more over (S, x):

$$\mathbb{E}_S\big[L_D(\hat{h})\big] - L_D^* \le 2\,\mathbb{E}_{S,x}|\hat{\eta}(x) - \eta(x)| \le 2\sqrt{\mathbb{E}_{S,x}\big[(\hat{\eta}(x) - \eta(x))^2\big]}.$$

As the left side is ≥ 0, it suffices to show $\mathbb{E}_{S,x}[(\hat{\eta}(x) - \eta(x))^2] \to 0$. We first assume that η is uniformly continuous (A1), which is not necessary but simplifies the argument. Introduce $\tilde{\eta}(x) = \sum_{i=1}^m \eta(x_i)\, w_i(x)$, the estimate with each label replaced by its conditional mean. By $(a+b)^2 \le 2a^2 + 2b^2$ (which is $(a-b)^2 \ge 0$),

$$(\hat{\eta} - \eta)^2 = \big((\hat{\eta} - \tilde{\eta}) + (\tilde{\eta} - \eta)\big)^2 \le 2(\hat{\eta} - \tilde{\eta})^2 + 2(\tilde{\eta} - \eta)^2,$$

and, using $\sum_i w_i(x) = 1$ to write $\eta(x) = \sum_i \eta(x) w_i(x)$,

$$\hat{\eta}(x) - \tilde{\eta}(x) = \sum_{i=1}^m \big(\mathbb{1}\{y_i = 1\} - \eta(x_i)\big)\, w_i(x), \qquad \tilde{\eta}(x) - \eta(x) = \sum_{i=1}^m \big(\eta(x_i) - \eta(x)\big)\, w_i(x).$$

We show that both squared terms tend to 0 in expectation.

Noise term. Let $\xi_i = \mathbb{1}\{y_i = 1\} - \eta(x_i)$. Given the inputs $x, x_1, \dots, x_m$, the weights are fixed and the ξ_i are independent, with $\mathbb{E}[\xi_i \mid \text{inputs}] = \eta(x_i) - \eta(x_i) = 0$ and $|\xi_i| \le 1$. Hence for $i \ne j$, by the tower property, $\mathbb{E}[\xi_i \xi_j w_i w_j] = \mathbb{E}\big[w_i w_j\, \mathbb{E}[\xi_i \mid \text{inputs}]\, \mathbb{E}[\xi_j \mid \text{inputs}]\big] = 0$. Dropping these cross terms, then using $\xi_i^2 \le 1$, $w_i^2 \le w_i \max_j w_j$ and $\sum_i w_i = 1$,

$$\begin{aligned}
\mathbb{E}_{S,x}\big[(\hat{\eta}(x) - \tilde{\eta}(x))^2\big] &= \mathbb{E}\Big[\sum_{i=1}^m \sum_{j=1}^m \xi_i \xi_j\, w_i(x) w_j(x)\Big] = \mathbb{E}\Big[\sum_{i=1}^m \xi_i^2\, w_i(x)^2\Big] \\
&\le \mathbb{E}\Big[\sum_{i=1}^m w_i(x)^2\Big] \le \mathbb{E}\Big[\max_{i \le m} w_i(x) \sum_{i=1}^m w_i(x)\Big] = \mathbb{E}\Big[\max_{i \le m} w_i(x)\Big] \to 0,
\end{aligned}$$

by condition (iii).

Bias term. Jensen's inequality for the convex $z \mapsto z^2$ with weights $w_i \ge 0$ summing to 1, $\big(\sum_i w_i z_i\big)^2 \le \sum_i w_i z_i^2$, gives

$$\mathbb{E}_{S,x}\big[(\tilde{\eta}(x) - \eta(x))^2\big] = \mathbb{E}\Big[\Big(\sum_{i=1}^m w_i(x)\big(\eta(x_i) - \eta(x)\big)\Big)^2\Big] \le \mathbb{E}\Big[\sum_{i=1}^m w_i(x)\big(\eta(x_i) - \eta(x)\big)^2\Big].$$

Fix $\epsilon > 0$. By (A1) there is $a_\epsilon > 0$ with $\|x_i - x\| \le a_\epsilon \Rightarrow |\eta(x_i) - \eta(x)| \le \epsilon$. Split each term with $1 = \mathbb{1}\{\|x_i - x\| \le a_\epsilon\} + \mathbb{1}\{\|x_i - x\| > a_\epsilon\}$, bounding $(\eta(x_i) - \eta(x))^2$ by ϵ^2 on the first event and by 1 on the second:

$$\begin{aligned}
\mathbb{E}\Big[\sum_{i=1}^m w_i(x)\big(\eta(x_i) - \eta(x)\big)^2\Big] &\le \epsilon^2\, \mathbb{E}\Big[\sum_{i=1}^m w_i(x)\Big] + \mathbb{E}\Big[\sum_{i=1}^m w_i(x)\, \mathbb{1}\{\|x_i - x\| > a_\epsilon\}\Big] \\
&= \epsilon^2 + \mathbb{E}\Big[\sum_{i=1}^m w_i(x)\, \mathbb{1}\{\|x_i - x\| > a_\epsilon\}\Big].
\end{aligned}$$

The last expectation tends to 0 by (ii) with $a = a_\epsilon$. Hence $\limsup_{m \to \infty} \mathbb{E}_{S,x}[(\tilde{\eta}(x) - \eta(x))^2] \le \epsilon^2$ for every $\epsilon > 0$, so the limit is 0. This proves Stone's theorem for uniformly continuous η, without using condition (i).

General η (sketch). Since η is bounded, for every $\epsilon > 0$ there is a uniformly continuous η^* with $\mathbb{E}_x[(\eta(x) - \eta^*(x))^2] < \epsilon$. Let $\tilde{\eta}^*(x) = \sum_i \eta^*(x_i) w_i(x)$ and split

$$\tilde{\eta} - \eta = (\tilde{\eta} - \tilde{\eta}^*) + (\tilde{\eta}^* - \eta^*) + (\eta^* - \eta).$$

The bias argument applied to η^* gives $\mathbb{E}_{S,x}[(\tilde{\eta}^* - \eta^*)^2] \to 0$, and $\mathbb{E}_x[(\eta^* - \eta)^2] < \epsilon$ by construction. For the first term, Jensen as above and condition (i) with $f = (\eta - \eta^*)^2$ give

$$\mathbb{E}_{S,x}\big[(\tilde{\eta}(x) - \tilde{\eta}^*(x))^2\big] \le \mathbb{E}\Big[\sum_{i=1}^{m} w_i(x)\big(\eta(x_i) - \eta^*(x_i)\big)^2\Big] \le c\,\mathbb{E}_x\big[(\eta(x) - \eta^*(x))^2\big] < c\,\epsilon.$$

With $(a+b+d)^2 \le 3(a^2+b^2+d^2)$, $\limsup_m \mathbb{E}_{S,x}[(\tilde{\eta} - \eta)^2] \le 3(c+1)\epsilon$ for every $\epsilon > 0$, so the bias term tends to 0 without assuming (A1).

4 ERM over a hypothesis class and uniform convergence

4.1 ERM over a hypothesis class

Restricting the search space. ERM over all functions can overfit. The predictor of example 1.4 **memorises** the sample, $h_{\text{mem}}(x) = 1$ if $(x, 1) \in S$ and $h_{\text{mem}}(x) = 0$ otherwise: it has $L_S(h_{\text{mem}}) = 0$, but its risk is large as soon as $\mathcal{X}$ is much larger than the sample (0.9 in that example). The identity $\mathbb{E}_S[L_S(h)] = L_D(h)$ (section 1) holds for a predictor h fixed before S is drawn, not for one built from S such as h_{mem}. The search space $\mathcal{Y}^{\mathcal{X}}$ contains functions complex enough to fit any sample, so ERM is restricted to a **hypothesis class** (or concept class) $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$, such as the linear classifiers or the **decision stumps** on $\mathcal{X} = \mathbb{R}$,

$$\mathcal{H} = \big\{ x \mapsto b \cdot \operatorname{sign}(t - x) \;:\; t \in \mathbb{R},\ b \in \{\pm 1\} \big\} \subseteq \{-1, +1\}^{\mathbb{R}},$$

which predict b left of the threshold t and $-b$ right of it.

Definition 4.1 (ERM over $\mathcal{H}$, approximation and estimation error)**. ERM over $\mathcal{H}$** returns a minimiser of the empirical risk over $\mathcal{H}$, $\hat{h} \in \arg\min_{h \in \mathcal{H}} L_S(h)$ (also written $\hat{h}_{\text{ERM}}$). The smallest risk in $\mathcal{H}$ is $L_D(\mathcal{H}) = \inf_{h \in \mathcal{H}} L_D(h)$, a minimum when $\mathcal{H}$ is finite. Since $\hat{h} \in \mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$, we have $L_D^* \le L_D(\mathcal{H}) \le L_D(\hat{h})$, and the excess risk of $\hat{h}$ splits into two terms ≥ 0,

$$L_D(\hat{h}) - L_D^* = \underbrace{L_D(\mathcal{H}) - L_D^*}_{\text{approximation error}} + \underbrace{L_D(\hat{h}) - L_D(\mathcal{H})}_{\text{estimation error}}.$$

The split only adds and subtracts $L_D(\mathcal{H})$, so it holds for every $\hat{h} \in \mathcal{H}$, e.g. for $\hat{h} = \arg\min_{h \in \mathcal{H}} J(h)$ with any cost function J (ERM is $J = L_S(\cdot)$). The **approximation error**, or **inductive bias**, is the price of the model assumption encoded by $\mathcal{H}$ (e.g. a linear decision boundary). It does not involve S, so no learning algorithm can reduce it, and it decreases as $\mathcal{H}$ grows, because an infimum over a larger set is smaller:

$$\mathcal{H} \subseteq \mathcal{H}' \implies L_D(\mathcal{H}) - L_D^* \ge L_D(\mathcal{H}') - L_D^*.$$

The **estimation error** compares $\hat{h}$ with the best predictor in $\mathcal{H}$. It depends on the learning algorithm, which can control it, and for ERM it typically increases with $\mathcal{H}$.

Underfitting and overfitting. Enlarging $\mathcal{H}$ (linear classifiers, then quadratic decision boundaries, and so on) admits more complex predictors: the approximation error and the training error $L_S(\hat{h})$ decrease, while the estimation error increases. The test error $L_D(\hat{h})$ is therefore large for a small $\mathcal{H}$, because of the approximation error (**underfitting**), and large for a large $\mathcal{H}$, because of the estimation error (**overfitting**); it is smallest in between (fig. 4). This is the bias-variance tradeoff of section 2, with the approximation error as bias.

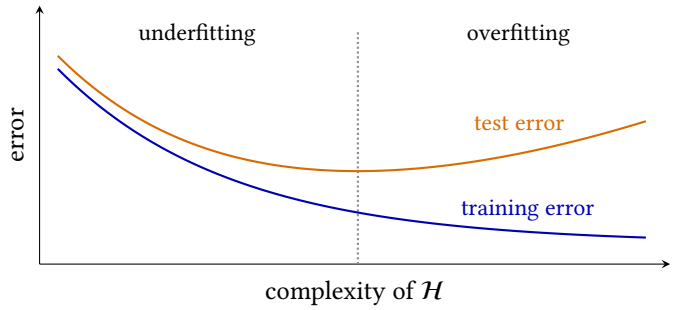


Figure 4: Training and test error as $\mathcal{H}$ grows.

4.2 Finite hypothesis classes

To complete the chain $L_D^* \le L_D(\mathcal{H}) \le L_D(\hat{h}) \le ?$ with an upper bound, in this subsection $\mathcal{H}$ is finite, ℓ is the 0-1 loss and $\hat{h}$ is ERM over $\mathcal{H}$. There are two settings: the **simple case** of deterministic labels $y = h_0(x)$ for some $h_0 \in \mathcal{H}$, and the general case of random labels $(x, y) \sim \mathcal{D}$, which also covers $y = h_0(x)$ with $h_0 \notin \mathcal{H}$.

Two probability tools. For events $A_1, \dots, A_n$, the **union bound** states $\mathbb{P}(A_1 \cup \cdots \cup A_n) \le \sum_{i=1}^n \mathbb{P}(A_i)$; if the events are independent, $\mathbb{P}(A_1 \cap \cdots \cap A_n) = \prod_{i=1}^n \mathbb{P}(A_i)$.

Theorem 4.2 (Tail bound for a maximum)**.** *For random variables $Z_1, \dots, Z_n$ and $\epsilon \in \mathbb{R}$,*

$$\mathbb{P}\Big(\max_{1 \le i \le n} Z_i > \epsilon \Big) \le \sum_{i=1}^n \mathbb{P}(Z_i > \epsilon).$$

Proof. The maximum exceeds ϵ exactly when some Z_i does, so $\{\max_i Z_i > \epsilon\} = \bigcup_{i=1}^n \{Z_i > \epsilon\}$; apply the union bound. □

Theorem 4.3 (Hoeffding's inequality)**.** *Let $Z_1, \dots, Z_m$ be independent random variables with $\mathbb{P}(Z_i \in [a_i, b_i]) = 1$ for all i. Then for every $\epsilon > 0$,*

$$\mathbb{P}\Big(\Big| \sum_{i=1}^m \big(Z_i - \mathbb{E}[Z_i]\big) \Big| > \epsilon \Big) \le 2 \exp\Big(- \frac{2\epsilon^2}{\sum_{i=1}^m (b_i - a_i)^2} \Big).$$

A one-sided version of this bound was used in the proof of lemma 3.5.

Simple case.

Theorem 4.4 (Generalisation error bound in the simple case)**.** *Let $\mathcal{H}$ be finite, ℓ the 0-1 loss, $y = h_0(x)$ with $h_0 \in \mathcal{H}$, and $\hat{h}$ ERM over $\mathcal{H}$. For $\epsilon \in (0, 1)$,*

$$\mathbb{P}_{S \sim \mathcal{D}^m}\big(L_D(\hat{h}) > \epsilon \big) \le |\mathcal{H}| e^{-m\epsilon}.$$

Equivalently, for $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$,

$$L_D(\hat{h}) \le \frac{\ln|\mathcal{H}| + \ln(1/\delta)}{m}.$$

Here $L_D(\mathcal{H}) = L_D(h_0) = 0$, and $L_S(\hat{h}) = 0$ because h_0 makes no training error and $\hat{h}$ minimises $L_S(\cdot)$. The two forms are linked by $\delta = |\mathcal{H}| e^{-m\epsilon}$: taking logarithms, $\ln \delta = \ln|\mathcal{H}| - m\epsilon$, i.e. $\epsilon = \frac{1}{m}\big(\ln|\mathcal{H}| + \ln(1/\delta) \big)$, and $\mathbb{P}(L_D(\hat{h}) > \epsilon) \le \delta$ is the same as $\mathbb{P}(L_D(\hat{h}) \le \epsilon) \ge 1 - \delta$. For instance, take $\delta = 0.01$, so that $\ln(1/\delta) \approx 4.6 < 5$. If the experiment "draw $S \sim \mathcal{D}^m$, run ERM, compute $L_D(\hat{h})$" is repeated many times, the bound $L_D(\hat{h}) \le \frac{1}{m}(\ln|\mathcal{H}| + 5)$ holds in (approximately at least) 99% of the runs; in practice there is a single run, and the bound typically holds for it.

Proof. We prove the first form. Let $\mathcal{H}_{\text{bad}} = \{h \in \mathcal{H} : L_D(h) > \epsilon\}$; it depends on $\mathcal{D}$ only, not on S. If $L_D(\hat{h}) > \epsilon$, then $\hat{h} \in \mathcal{H}_{\text{bad}}$ and $L_S(\hat{h}) = 0$, so some $h \in \mathcal{H}_{\text{bad}}$ has $L_S(h) = 0$:

$$\big\{L_D(\hat{h}) > \epsilon\big\} \subseteq \bigcup_{h \in \mathcal{H}_{\text{bad}}} \big\{L_S(h) = 0\big\}.$$

Fix $h \in \mathcal{H}_{\text{bad}}$. The losses are ≥ 0, so $L_S(h) = 0$ exactly when $\ell(h(x_i), y_i) = 0$ for every i. Each $\ell(h(x_i), y_i)$ is a Bernoulli variable with $\mathbb{P}\big(\ell(h(x_i), y_i) = 1\big) = \mathbb{P}_{(x,y)\sim D}(h(x) \neq y) = L_D(h) > \epsilon$, so $\mathbb{P}\big(\ell(h(x_i), y_i) = 0\big) \leq 1 - \epsilon$. Hence

$$\begin{aligned}
\mathbb{P}_{S\sim\mathcal{D}^m}\big(L_D(\hat{h}) > \epsilon\big) &\leq \sum_{h \in \mathcal{H}_{\text{bad}}} \mathbb{P}_{S\sim\mathcal{D}^m}\big(L_S(h) = 0\big) && \text{(union bound)} \\
&= \sum_{h \in \mathcal{H}_{\text{bad}}} \prod_{i=1}^{m} \mathbb{P}_{(x_i,y_i)\sim D}\big(\ell(h(x_i), y_i) = 0\big) && \text{(pairs independent, } h \text{ fixed)} \\
&\leq \sum_{h \in \mathcal{H}_{\text{bad}}} (1-\epsilon)^m = |\mathcal{H}_{\text{bad}}|(1-\epsilon)^m \\
&\leq |\mathcal{H}|(1-\epsilon)^m \leq |\mathcal{H}|e^{-m\epsilon} && (\mathcal{H}_{\text{bad}} \subseteq \mathcal{H},\ 1-\epsilon \leq e^{-\epsilon}).
\end{aligned}$$

The last step uses that the convex function $e^{-\epsilon}$ lies above its tangent $1 - \epsilon$ at 0, raised to the power m (both sides are ≥ 0). □

General case: uniform convergence.

Theorem 4.5 (Uniform convergence of $L_S(\cdot)$ for finite $\mathcal{H}$). *Let $\mathcal{H} \subseteq \{-1, +1\}^{\mathcal{X}}$ be finite, ℓ the 0-1 loss and $\epsilon \in (0, 1)$. Then*

$$\mathbb{P}_{S\sim\mathcal{D}^m}\Big(\max_{h \in \mathcal{H}} |L_S(h) - L_D(h)| > \epsilon\Big) \leq 2|\mathcal{H}|e^{-2m\epsilon^2}.$$

Equivalently, for $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\max_{h \in \mathcal{H}} |L_S(h) - L_D(h)| \leq \sqrt{\frac{\ln|\mathcal{H}| + \ln(2/\delta)}{2m}}.$$

As before, the forms are linked by $\delta = 2|\mathcal{H}|e^{-2m\epsilon^2}$, i.e. $2m\epsilon^2 = \ln(2|\mathcal{H}|/\delta) = \ln|\mathcal{H}| + \ln(2/\delta)$.

Proof. By theorem 4.2 applied to the $|\mathcal{H}|$ variables $|L_S(h) - L_D(h)|$,

$$\mathbb{P}\Big(\max_{h \in \mathcal{H}} |L_S(h) - L_D(h)| > \epsilon\Big) \leq \sum_{h \in \mathcal{H}} \mathbb{P}\big(|L_S(h) - L_D(h)| > \epsilon\big).$$

Fix $h \in \mathcal{H}$ and let $Z_i = \ell(h(x_i), y_i)$. The Z_i are independent (the pairs are), take values in $\{0, 1\} \subseteq [0, 1]$, and $\mathbb{E}[Z_i] = L_D(h)$ because $(x_i, y_i) \sim \mathcal{D}$. So $L_S(h) - L_D(h) = \frac{1}{m}\sum_{i=1}^{m}(Z_i - \mathbb{E}[Z_i])$, and Hoeffding's inequality (theorem 4.3) with $a_i = 0$, $b_i = 1$, $\sum_i (b_i - a_i)^2 = m$, applied with $m\epsilon$ in place of ϵ, gives

$$\mathbb{P}\big(|L_S(h) - L_D(h)| > \epsilon\big) = \mathbb{P}\Big(\Big|\sum_{i=1}^{m}(Z_i - \mathbb{E}[Z_i])\Big| > m\epsilon\Big) \leq 2\exp\Big(-\frac{2m^2\epsilon^2}{m}\Big) = 2e^{-2m\epsilon^2}.$$

Summing over the $|\mathcal{H}|$ hypotheses gives the claim. □

Corollary 4.6 (Generalisation error bound for finite $\mathcal{H}$). *Under the assumptions of theorem 4.5, let $\hat{h}$ be ERM over $\mathcal{H}$. For $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$L_D(\hat{h}) \leq L_D(\mathcal{H}) + \sqrt{\frac{2\ln|\mathcal{H}| + 2\ln(2/\delta)}{m}}.$$

Proof. Let $\epsilon_\delta = \sqrt{(\ln|\mathcal{H}| + \ln(2/\delta))/(2m)}$. By theorem 4.5, with probability at least $1-\delta$ we have $|L_S(h) - L_D(h)| \le \epsilon_\delta$ simultaneously for all $h \in \mathcal{H}$, in particular $L_D(\hat{h}) - L_S(\hat{h}) \le \epsilon_\delta$ and $L_S(h) \le L_D(h) + \epsilon_\delta$ for every $h \in \mathcal{H}$. On this event,

$$\begin{aligned} L_D(\hat{h}) &\le L_S(\hat{h}) + \epsilon_\delta = \min_{h\in\mathcal{H}} L_S(h) + \epsilon_\delta && (\hat{h} \text{ minimises } L_S(\cdot)) \\ &\le \min_{h\in\mathcal{H}} L_D(h) + 2\epsilon_\delta = L_D(\mathcal{H}) + \sqrt{\frac{4(\ln|\mathcal{H}| + \ln(2/\delta))}{2m}} && (L_S(h) \le L_D(h) + \epsilon_\delta \text{ for all } h), \end{aligned}$$

which is the claim. □

The bound applies to $\hat{h}$ although $\hat{h}$ depends on S, because it holds for all $h \in \mathcal{H}$ at once; this is why the deviation $L_D(h) - L_S(h)$ is bounded uniformly over $\mathcal{H}$.

Bound in expectation. The bounds above hold with high probability, whereas the risks of OLS (theorem 2.4) and of k-NN (theorem 3.9) were studied in expectation over S. Uniform convergence also gives a bound of this kind.

Theorem 4.7 (Bound on the expected risk of ERM)**.** *Let $\mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$ be finite, ℓ the 0-1 loss and $\hat{h}$ ERM over $\mathcal{H}$. Then*

$$\mathbb{E}_S\big[L_D(\hat{h})\big] \le L_D(\mathcal{H}) + \mathbb{E}_S\Big[\max_{h\in\mathcal{H}} \big(L_D(h) - L_S(h)\big)\Big] \le L_D(\mathcal{H}) + \sqrt{\frac{2\ln|\mathcal{H}| + \ln(8m)}{m}}.$$

Proof. First inequality. Let $h_0 \in \arg\min_{h\in\mathcal{H}} L_D(h)$, so that $L_D(h_0) = L_D(\mathcal{H})$. Adding and subtracting $L_S(\hat{h})$ and $L_S(h_0)$,

$$L_D(\hat{h}) = \big(L_D(\hat{h}) - L_S(\hat{h})\big) + \underbrace{\big(L_S(\hat{h}) - L_S(h_0)\big)}_{\le 0 \ (\hat{h} \text{ minimises } L_S(\cdot))} + L_S(h_0) \le \max_{h\in\mathcal{H}} \big(L_D(h) - L_S(h)\big) + L_S(h_0).$$

Take $\mathbb{E}_S$: since h_0 does not depend on S, $\mathbb{E}_S[L_S(h_0)] = L_D(h_0) = L_D(\mathcal{H})$.

Second inequality. Let $Z_S = \max_{h\in\mathcal{H}}(L_D(h) - L_S(h))$. For the 0-1 loss $L_D(h) \le 1$ and $L_S(h) \ge 0$, so $Z_S \le 1$. For any $\epsilon > 0$ exactly one of $\mathbb{1}\{Z_S \le \epsilon\}$, $\mathbb{1}\{Z_S > \epsilon\}$ equals 1, and on the first event $Z_S \le \epsilon$, on the second $Z_S \le 1$; hence

$$Z_S \le \epsilon \cdot \mathbb{1}\{Z_S \le \epsilon\} + 1 \cdot \mathbb{1}\{Z_S > \epsilon\}, \qquad \text{so} \qquad \mathbb{E}_S[Z_S] \le \epsilon\, \mathbb{P}(Z_S \le \epsilon) + \mathbb{P}(Z_S > \epsilon) \le \epsilon + \mathbb{P}(Z_S > \epsilon).$$

Choose

$$\delta = \frac{1}{\sqrt{2m}} \in (0,1), \qquad \epsilon = \sqrt{\frac{\ln|\mathcal{H}| + \ln(2/\delta)}{2m}}, \qquad \text{where} \quad \ln\frac{2}{\delta} = \ln\big(2\sqrt{2m}\big) = \frac{1}{2}\ln(8m).$$

Since $Z_S \le \max_{h\in\mathcal{H}} |L_S(h) - L_D(h)|$, theorem 4.5 gives $\mathbb{P}(Z_S > \epsilon) \le \delta$. Moreover $\delta < \epsilon$: indeed $\ln|\mathcal{H}| \ge 0$ and $\frac{1}{2}\ln(8m) \ge \frac{1}{2}\ln 8 > 1$ for $m \ge 1$, so $\epsilon^2 > \frac{1}{2m} = \delta^2$. Therefore

$$\mathbb{E}_S[Z_S] \le \epsilon + \delta \le 2\epsilon = \sqrt{\frac{4\ln|\mathcal{H}| + 2\ln(8m)}{2m}} = \sqrt{\frac{2\ln|\mathcal{H}| + \ln(8m)}{m}}.$$ □

Summary for finite $\mathcal{H}$. For the 0-1 loss and $\hat{h}$ ERM over a finite $\mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$:

Result	Guarantee	Bound
Theorem 4.4 (simple case)	probability $\geq 1-\delta$	$L_D(\hat{h}) \leq \dfrac{\ln\lvert\mathcal{H}\rvert + \ln(1/\delta)}{m}$
Theorem 4.5	probability $\geq 1-\delta$	$\max_{h\in\mathcal{H}} \lvert L_S(h) - L_D(h)\rvert \leq \sqrt{\dfrac{\ln\lvert\mathcal{H}\rvert + \ln(2/\delta)}{2m}}$
Corollary 4.6	probability $\geq 1-\delta$	$L_D(\hat{h}) \leq L_D(\mathcal{H}) + \sqrt{\dfrac{2\ln\lvert\mathcal{H}\rvert + 2\ln(2/\delta)}{m}}$
Theorem 4.7	in expectation	$\mathbb{E}_S\big[L_D(\hat{h})\big] \leq L_D(\mathcal{H}) + \sqrt{\dfrac{2\ln\lvert\mathcal{H}\rvert + \ln(8m)}{m}}$

4.3 Rademacher complexity

For an infinite $\mathcal{H}$ the bounds above are void ($\ln|\mathcal{H}| = \infty$). The idea of **symmetrisation** is to compare the training sample S with a second sample $S' \sim D^m$ independent of S: if $\sup_{h\in\mathcal{H}} |L_{S'}(h) - L_S(h)|$ is small, then $\sup_{h\in\mathcal{H}} |L_D(h) - L_S(h)|$ is small as well. Random signs, which swap points between the two samples, turn this comparison into the following quantity.

Definition 4.8 (Rademacher complexity)**.** Let $\mathcal{Z}$ be a set, $\mathcal{F} \subseteq \mathbb{R}^{\mathcal{Z}}$ a class of real-valued functions on $\mathcal{Z}$, and $Z = \{z_1, \dots, z_m\} \subseteq \mathcal{Z}$. **Rademacher variables** $\sigma_1, \dots, \sigma_m$ are i.i.d. uniform on $\{-1, +1\}$. The **empirical Rademacher complexity** of $\mathcal{F}$ with respect to Z is

$$R_Z(\mathcal{F}) = \mathbb{E}_{\sigma_1,\dots,\sigma_m}\Big[\sup_{f\in\mathcal{F}} \frac{1}{m}\sum_{i=1}^{m} \sigma_i\, f(z_i)\Big].$$

For a distribution D on $\mathcal{Z}$, the (distribution-dependent) **Rademacher complexity** is $R_{D,m}(\mathcal{F}) = \mathbb{E}_{Z\sim D^m}\big[R_Z(\mathcal{F})\big]$.

Examples are a hypothesis class $\mathcal{F} = \mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$ on $\mathcal{Z} = \mathcal{X}$, the linear functions $\mathcal{F} = \{z \mapsto w^\top z : w \in \mathbb{R}^p\}$ on $\mathcal{Z} = \mathbb{R}^p$, and the **loss class** $\mathcal{F} = \ell \circ \mathcal{H} = \{(x, y) \mapsto \ell(h(x), y) : h \in \mathcal{H}\}$ on $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$. For learning, the last one matters: with $Z = S$ we write $R_S(\ell \circ \mathcal{H})$, and

$$R_{D,m} = R_{D,m}(\ell \circ \mathcal{H}) = \mathbb{E}_{S\sim D^m}\, \mathbb{E}_{\sigma_1,\dots,\sigma_m}\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^{m} \sigma_i\, \ell(h(x_i), y_i)\Big].$$

Intuition. For $\mathcal{F} = \mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$ and $Z = \{x_1, \dots, x_m\}$, view the σ_i as random labels of the x_i. The quantity $\frac{1}{m}\sum_i \sigma_i h(x_i)$ measures how well h aligns with these labels: since $\sigma_i h(x_i)$ is $+1$ if $h(x_i) = \sigma_i$ and -1 otherwise, it equals $1 - 2\cdot\frac{1}{m}\#\{i : h(x_i) \neq \sigma_i\}$. So $R_{x_1,\dots,x_m}(\mathcal{H})$ is the expected best alignment with random labels that $\mathcal{H}$ can achieve: it is large when $\mathcal{H}$ can fit pure noise.

For $\mathcal{F} = \ell \circ \mathcal{H}$ and $Z = S$, the signs split the sample at random into $S_+ = \{(x_i, y_i) : \sigma_i = +1\}$ and $S_- = \{(x_i, y_i) : \sigma_i = -1\}$, and

$$\frac{1}{m}\sum_{i=1}^{m} \sigma_i\, \ell(h(x_i), y_i) = \frac{1}{m}\sum_{(x_i,y_i)\in S_+} \ell(h(x_i), y_i) - \frac{1}{m}\sum_{(x_i,y_i)\in S_-} \ell(h(x_i), y_i) = \frac{|S_+|}{m} L_{S_+}(h) - \frac{|S_-|}{m} L_{S_-}(h).$$

For roughly equal halves, $|S_+| \approx |S_-| \approx \frac{m}{2}$, this gives $R_S(\ell \circ \mathcal{H}) \approx \frac{1}{2}\mathbb{E}_\sigma\big[\sup_{h\in\mathcal{H}} \big(L_{S_+}(h) - L_{S_-}(h)\big)\big]$: the Rademacher complexity measures how much the empirical risks on two random halves of the sample can disagree, which is the comparison behind symmetrisation.

Symmetrisation and risk bounds.

Theorem 4.9 (Symmetrisation bound in expectation)**.** *For any label set $\mathcal{Y}$, hypothesis class $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$ and loss $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$,*

$$\mathbb{E}_{S\sim D^m}\Big[\sup_{h\in\mathcal{H}}\big(L_D(h) - L_S(h)\big)\Big] \le 2R_{D,m}, \qquad \mathbb{E}_{S\sim D^m}\Big[\sup_{h\in\mathcal{H}}\big(L_S(h) - L_D(h)\big)\Big] \le 2R_{D,m}.$$

The proof is in section 4.4.

Theorem 4.10 (Rademacher bound on the expected risk)**.** *Let $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$ be any (possibly infinite) hypothesis class, ℓ any loss, and $\hat{h}$ ERM over $\mathcal{H}$ based on $S \sim D^m$. With $L_D(\mathcal{H}) = \inf_{h\in\mathcal{H}} L_D(h)$,*

$$\mathbb{E}_S\big[L_D(\hat{h})\big] \le L_D(\mathcal{H}) + 2R_{D,m}.$$

Proof. The proof of the first inequality of theorem 4.7 used a minimiser h_0 of $L_D(\cdot)$ over $\mathcal{H}$, which may not exist now: an infimum over an infinite $\mathcal{H}$ need not be attained. Instead fix an arbitrary $h_0 \in \mathcal{H}$. As there, $L_S(\hat{h}) \le L_S(h_0)$ gives

$$L_D(\hat{h}) = \big(L_D(\hat{h}) - L_S(\hat{h})\big) + \big(L_S(\hat{h}) - L_S(h_0)\big) + L_S(h_0) \le \sup_{h\in\mathcal{H}}\big(L_D(h) - L_S(h)\big) + L_S(h_0).$$

Taking $\mathbb{E}_S$, with $\mathbb{E}_S[L_S(h_0)] = L_D(h_0)$ (h_0 is fixed) and theorem 4.9,

$$\mathbb{E}_S\big[L_D(\hat{h})\big] \le 2R_{D,m} + L_D(h_0).$$

The left side does not depend on h_0, so taking the infimum over $h_0 \in \mathcal{H}$ gives the claim. □

Theorem 4.11 (Generalisation bound using Rademacher complexity)**.** *Let $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$ be a hypothesis class (not necessarily binary), $\hat{h}$ ERM over $\mathcal{H}$, and ℓ a loss with $0 \le \ell(h(x), y) \le c$ for all $h \in \mathcal{H}$ and $(x, y) \in \mathcal{X} \times \mathcal{Y}$. For every distribution D and $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $S \sim D^m$,*

$$L_D(\hat{h}) - L_S(\hat{h}) \le \sup_{h\in\mathcal{H}}\big(L_D(h) - L_S(h)\big) \le 2R_{D,m}(\ell \circ \mathcal{H}) + c\sqrt{\frac{\ln(1/\delta)}{2m}},$$

and, with probability at least $1 - \delta$,

$$L_D(\hat{h}) - L_S(\hat{h}) \le \sup_{h\in\mathcal{H}}\big(L_D(h) - L_S(h)\big) \le 2R_S(\ell \circ \mathcal{H}) + 3c\sqrt{\frac{\ln(2/\delta)}{2m}}.$$

In both, the first inequality holds because $\hat{h} \in \mathcal{H}$. The second bound involves the empirical $R_S(\ell \circ \mathcal{H})$, which depends on the sample only and not on D. The proof is omitted (Bach, Learning Theory from First Principles, p. 93).

4.4 Proof of the symmetrisation bound (not examinable)

This subsection is not part of the exam. We prove theorem 4.9. Let $S' = ((x_i', y_i'))_{i=1}^m \sim D^m$ be an independent copy of $S = ((x_i, y_i))_{i=1}^m$, write $\mathbb{E}_{S'|S}$ for the expectation over S' given S, and abbreviate

$$\ell_i(h) = \ell(h(x_i), y_i), \qquad \ell_i'(h) = \ell(h(x_i'), y_i').$$

We use four facts.

(i) For a class $\mathcal{F}$ of real-valued functions and a random variable z, $\sup_{f\in\mathcal{F}} \mathbb{E}_z[f(z)] \le \mathbb{E}_z\big[\sup_{f\in\mathcal{F}} f(z)\big]$. Indeed, for every $f \in \mathcal{F}$, $f(z) \le \sup_{g\in\mathcal{F}} g(z)$, so $\mathbb{E}_z[f(z)] \le \mathbb{E}_z[\sup_{g\in\mathcal{F}} g(z)]$; the right side does not depend on f, so the supremum over f satisfies the same bound.

(ii) Since S' is independent of S, $\mathbb{E}_{S'|S}[\ell_i'(h)] = \mathbb{E}_{S'}[\ell_i'(h)] = L_D(h)$, hence $L_D(h) = \mathbb{E}_{S'|S}\big[\frac{1}{m}\sum_{i=1}^m \ell_i'(h)\big]$.

(iii) Given S, the training pairs are fixed: $\mathbb{E}_{S'|S}[\ell_i(h)] = \ell_i(h)$, hence $L_S(h) = \mathbb{E}_{S'|S}\big[\frac{1}{m}\sum_{i=1}^m \ell_i(h)\big]$.

(iv) Fix signs $\sigma \in \{\pm 1\}^m$ and let $(\tilde{S}, \tilde{S}')$ be obtained from (S, S') by swapping (x_i, y_i) with (x_i', y_i') for every i with $\sigma_i = -1$. All $2m$ pairs are i.i.d. from $\mathcal{D}$, so $(\tilde{S}, \tilde{S}')$ has the same law as (S, S'). The difference $\ell_i'(h) - \ell_i(h)$ computed on $(\tilde{S}, \tilde{S}')$ equals $\sigma_i\big(\ell_i'(h) - \ell_i(h)\big)$ computed on (S, S'), since it changes sign exactly at the swapped indices. Hence, for every fixed σ,

$$\mathbb{E}_{S,S'}\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \sigma_i\big(\ell_i'(h) - \ell_i(h)\big)\Big] = \mathbb{E}_{S,S'}\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \big(\ell_i'(h) - \ell_i(h)\big)\Big],$$

and the same holds after averaging over Rademacher variables $\sigma_1, \dots, \sigma_m$ independent of S, S'.

Proof of theorem 4.9. By (ii) and (iii) and linearity of $\mathbb{E}_{S'|S}$,

$$\begin{aligned}
\mathbb{E}_S\Big[\sup_{h\in\mathcal{H}} \big(L_{\mathcal{D}}(h) - L_S(h)\big)\Big] &= \mathbb{E}_S\Big[\sup_{h\in\mathcal{H}} \mathbb{E}_{S'|S}\Big[\frac{1}{m}\sum_{i=1}^m \big(\ell_i'(h) - \ell_i(h)\big)\Big]\Big] \\
&\le \mathbb{E}_S\Big[\mathbb{E}_{S'|S}\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \big(\ell_i'(h) - \ell_i(h)\big)\Big]\Big] && \text{((i) given } S\text{, with } z = S') \\
&= \mathbb{E}_{S,S'}\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \big(\ell_i'(h) - \ell_i(h)\big)\Big] && \text{(tower property)} \\
&= \mathbb{E}_{S,S',\sigma}\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \sigma_i\big(\ell_i'(h) - \ell_i(h)\big)\Big] && \text{(iv)} \\
&\le \mathbb{E}_{S',\sigma}\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \sigma_i\, \ell_i'(h)\Big] \\
&\quad + \mathbb{E}_{S,\sigma}\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m (-\sigma_i)\, \ell_i(h)\Big],
\end{aligned}$$

where the last step uses $\sup_h(a(h)+b(h)) \le \sup_h a(h) + \sup_h b(h)$. The first term is $R_{D,m}$ by definition, as $S' \sim \mathcal{D}^m$. In the second, $(-\sigma_1, \dots, -\sigma_m)$ has the same law as $(\sigma_1, \dots, \sigma_m)$, so it is $R_{D,m}$ as well; this proves the first bound. The second bound follows in the same way with $\ell_i(h) - \ell_i'(h)$ in place of $\ell_i'(h) - \ell_i(h)$. □

5 VC dimension and error bounds for the 0-1 loss

The bounds of theorems 4.10 and 4.11 control the risk of ERM through the Rademacher complexity $R_{D,m}(\ell \circ \mathcal{H})$ or $R_S(\ell \circ \mathcal{H})$. For losses that take finitely many values, such as the 0-1 loss, this section bounds it by quantities that depend on $\mathcal{H}$ only, not on S or $\mathcal{D}$. Massart's lemma bounds the Rademacher complexity of a finite set; since on m points such a loss produces finitely many loss vectors, it applies to infinite $\mathcal{H}$ as well, through the growth function; the VC dimension measures the complexity of $\mathcal{H}$ and bounds the growth function (Sauer's lemma).

5.1 Massart's lemma and the growth function

Rademacher complexity of a finite set. For $\mathcal{F} \subseteq \mathbb{R}^m$ let

$$R(\mathcal{F}) = \mathbb{E}_{\sigma_1,\dots,\sigma_m}\Big[\sup_{f\in\mathcal{F}} \frac{1}{m}\sum_{i=1}^m \sigma_i f_i\Big],$$

with Rademacher variables $\sigma_1, \dots, \sigma_m$. This is the empirical Rademacher complexity of definition 4.8 written through the vectors of values on the sample: for a hypothesis class $\mathcal{H}$, the 0-1 loss and a sample S, $R_S(\ell \circ \mathcal{H}) = R(\mathcal{F})$ with

$$\mathcal{F} = \{f^h : h \in \mathcal{H}\}, \qquad f^h = \big(\mathbb{1}\{h(x_1) \ne y_1\}, \dots, \mathbb{1}\{h(x_m) \ne y_m\}\big) \in \mathbb{R}^m.$$

Theorem 5.1 (Massart's finite class lemma). *Let $\mathcal{F} \subseteq \mathbb{R}^m$ be finite, with $\|f\| = \left(\sum_{i=1}^m f_i^2\right)^{1/2} \le r$ for every $f \in \mathcal{F}$. Then*

$$R(\mathcal{F}) \le \frac{r\sqrt{2\ln|\mathcal{F}|}}{m}.$$

The proof bounds the exponential of the supremum: for $\lambda > 0$, $e^{\lambda \sup_f a_f} = \sup_f e^{\lambda a_f} \le \sum_f e^{\lambda a_f}$, and for large λ the largest term dominates the sum, so the last step loses little; λ is chosen at the end to make the bound tight. It uses the following fact.

Lemma 5.2 (Hoeffding's lemma). *If a random variable z takes values in $[a, b]$, then $\mathbb{E}\left[e^{s(z-\mathbb{E}[z])}\right] \le e^{s^2(b-a)^2/8}$ for every $s \in \mathbb{R}$.*

Proof of theorem 5.1. Fix $\lambda > 0$. Since ln and exp are inverse to each other,

$$R(\mathcal{F}) = \frac{1}{\lambda} \ln \exp\left(\frac{\lambda}{m}\,\mathbb{E}_\sigma\left[\sup_{f\in\mathcal{F}} \sum_{i=1}^m \sigma_i f_i\right]\right).$$

We bound the exponential:

$$\begin{aligned}
\exp\left(\frac{\lambda}{m}\,\mathbb{E}_\sigma\left[\sup_{f\in\mathcal{F}} \sum_i \sigma_i f_i\right]\right) &\le \mathbb{E}_\sigma\left[\exp\left(\frac{\lambda}{m}\sup_{f\in\mathcal{F}} \sum_i \sigma_i f_i\right)\right] && \text{(Jensen, exp convex)}\\
&= \mathbb{E}_\sigma\left[\sup_{f\in\mathcal{F}} \exp\left(\frac{\lambda}{m}\sum_i \sigma_i f_i\right)\right] && \text{(exp increasing)}\\
&\le \sum_{f\in\mathcal{F}} \mathbb{E}_\sigma\left[\exp\left(\frac{\lambda}{m}\sum_i \sigma_i f_i\right)\right] && \text{(sup} \le \text{sum of terms} > 0\text{, linearity)}\\
&= \sum_{f\in\mathcal{F}} \prod_{i=1}^m \mathbb{E}_{\sigma_i}\left[\exp\left(\frac{\lambda f_i}{m}\,\sigma_i\right)\right] && (\sigma_i \text{ independent})\\
&\le \sum_{f\in\mathcal{F}} \prod_{i=1}^m \exp\left(\frac{\lambda^2 f_i^2}{2m^2}\right) && \text{(lemma 5.2)}\\
&= \sum_{f\in\mathcal{F}} \exp\left(\frac{\lambda^2\|f\|^2}{2m^2}\right) \le |\mathcal{F}| \exp\left(\frac{\lambda^2 r^2}{2m^2}\right) && (\|f\| \le r).
\end{aligned}$$

Lemma 5.2 is applied with $z = \sigma_i \in [-1, 1]$, $\mathbb{E}[\sigma_i] = 0$ and $s = \lambda f_i/m$, so that $s^2(b-a)^2/8 = \lambda^2 f_i^2 \cdot 4/(8m^2)$. Since ln is increasing,

$$R(\mathcal{F}) \le \frac{1}{\lambda} \ln\left(|\mathcal{F}| \exp\left(\frac{\lambda^2 r^2}{2m^2}\right)\right) = \frac{\ln|\mathcal{F}|}{\lambda} + \frac{\lambda r^2}{2m^2} =: g(\lambda) \qquad \text{for every } \lambda > 0.$$

Setting $g'(\lambda) = -\frac{\ln|\mathcal{F}|}{\lambda^2} + \frac{r^2}{2m^2} = 0$ gives the minimiser $\lambda^* = \frac{m}{r}\sqrt{2\ln|\mathcal{F}|}$, at which both terms of g equal $\frac{r}{2m}\sqrt{2\ln|\mathcal{F}|}$; hence $R(\mathcal{F}) \le g(\lambda^*) = \frac{r}{m}\sqrt{2\ln|\mathcal{F}|}$. □

Finite hypothesis classes. Let $\mathcal{H}$ be finite and ℓ the 0-1 loss. The vectors f^h have entries in $\{0, 1\}$, so $\|f^h\| \le \sqrt{m} =: r$, and $|\mathcal{F}| \le |\mathcal{H}|$. By theorem 5.1, for every sample S of size m,

$$R_S(\ell \circ \mathcal{H}) \le \frac{\sqrt{m}\sqrt{2\ln|\mathcal{H}|}}{m} = \sqrt{\frac{2\ln|\mathcal{H}|}{m}},$$

hence also $R_{D,m}(\ell \circ \mathcal{H}) = \mathbb{E}_S[R_S(\ell \circ \mathcal{H})] \le \sqrt{2\ln|\mathcal{H}|/m}$, and theorem 4.10 gives, for ERM over $\mathcal{H}$,

$$\mathbb{E}_S\left[L_D(\hat{h})\right] \le L_D(\mathcal{H}) + 2\sqrt{\frac{2\ln|\mathcal{H}|}{m}}.$$

A better constant follows by working with the labellings that $\mathcal{H}$ produces instead of the loss vectors.

Restriction to a sample. For $C = (x_1, \dots, x_m) \in \mathcal{X}^m$ (features only, no labels), the **restriction** of $\mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$ to C is the set of labellings of C produced by $\mathcal{H}$,

$$\mathcal{H}_{|C} = \big\{ (h(x_1), \dots, h(x_m)) \,:\, h \in \mathcal{H} \big\} \subseteq \{\pm 1\}^m.$$

It is finite even when $\mathcal{H}$ is not, $|\mathcal{H}_{|C}| \le \min\{|\mathcal{H}|, 2^m\}$, and its vectors have norm $\sqrt{m}$, so theorem 5.1 gives $R(\mathcal{H}_{|C}) \le \sqrt{2\ln|\mathcal{H}_{|C}|/m}$. For a sample S, $\mathcal{H}_{|S}$ is the restriction to its features $(x_1, \dots, x_m)$.

Lemma 5.3 (0-1 loss class and restriction). *For $\mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$, the 0-1 loss and every sample S, $R_S(\ell \circ \mathcal{H}) = \frac{1}{2} R(\mathcal{H}_{|S})$.*

Proof. For $y, y' \in \{\pm 1\}$, $yy' = 1$ if $y = y'$ and $yy' = -1$ otherwise, so $\mathbb{1}\{y \neq y'\} = \frac{1 - yy'}{2}$. Hence

$$\frac{1}{m} \sum_{i=1}^{m} \sigma_i \mathbb{1}\{h(x_i) \neq y_i\} = \frac{1}{2m} \sum_{i=1}^{m} \sigma_i + \frac{1}{2} \cdot \frac{1}{m} \sum_{i=1}^{m} (-\sigma_i y_i)\, h(x_i).$$

The first term does not depend on h, so it can be taken out of $\sup_{h \in \mathcal{H}}$, and its expectation over σ is 0. For fixed labels $y_i \in \{\pm 1\}$, the variables $\sigma'_i = -\sigma_i y_i$ are again i.i.d. uniform on $\{\pm 1\}$ (multiplying by $-y_i$ is a bijection of $\{\pm 1\}$). Therefore

$$R_S(\ell \circ \mathcal{H}) = \frac{1}{2}\, \mathbb{E}_{\sigma'} \Big[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma'_i\, h(x_i) \Big] = \frac{1}{2} R(\mathcal{H}_{|S}),$$

because $(h(x_1), \dots, h(x_m))$ ranges over $\mathcal{H}_{|S}$ as h ranges over $\mathcal{H}$. □

Corollary 5.4 (Improved bound for finite $\mathcal{H}$). *For a finite $\mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$, the 0-1 loss and $\hat{h}$ ERM over $\mathcal{H}$,*

$$\mathbb{E}_S\big[L_D(\hat{h})\big] \le L_D(\mathcal{H}) + \sqrt{\frac{2\ln|\mathcal{H}|}{m}}.$$

Proof. By lemma 5.3 and theorem 5.1 with $|\mathcal{H}_{|S}| \le |\mathcal{H}|$, $R_S(\ell \circ \mathcal{H}) = \frac{1}{2} R(\mathcal{H}_{|S}) \le \frac{1}{2}\sqrt{2\ln|\mathcal{H}|/m}$ for every S. Take $\mathbb{E}_S$ and apply theorem 4.10. □

This improves both the extra term $2\sqrt{2\ln|\mathcal{H}|/m}$ obtained above and the term $\sqrt{(2\ln|\mathcal{H}| + \ln(8m))/m}$ of theorem 4.7.

Growth function.

Definition 5.5 (Growth function). The **growth function** of $\mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$ is

$$\tau_{\mathcal{H}}(m) = \max_{C \subseteq \mathcal{X} \,:\, |C| = m} |\mathcal{H}_{|C}|,$$

the largest number of labellings of m points of $\mathcal{X}$ that $\mathcal{H}$ can produce.

In particular $R(\mathcal{H}_{|C}) \le \sqrt{2\ln \tau_{\mathcal{H}}(m)/m}$ for every C of size m.

Theorem 5.6 (Generalisation error bound in terms of the growth function). *Let ℓ be the 0-1 loss, $\mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$ any hypothesis class, $\mathcal{D}$ any distribution on $\mathcal{X} \times \{\pm 1\}$, $S \sim \mathcal{D}^m$ and $\hat{h}$ ERM over $\mathcal{H}$. Then*

$$\mathbb{E}_S\big[L_D(\hat{h}) - L_S(\hat{h})\big] \le \sqrt{\frac{2\ln \tau_{\mathcal{H}}(m)}{m}} \qquad \text{and} \qquad \mathbb{E}_S\big[L_D(\hat{h})\big] \le L_D(\mathcal{H}) + \sqrt{\frac{2\ln \tau_{\mathcal{H}}(m)}{m}}.$$

Proof. For every S, lemma 5.3 and $R(\mathcal{H}_{|S}) \le \sqrt{2\ln \tau_{\mathcal{H}}(m)/m}$ give $R_S(\ell \circ \mathcal{H}) \le \frac{1}{2}\sqrt{2\ln \tau_{\mathcal{H}}(m)/m}$, hence the same bound for $R_{D,m}(\ell \circ \mathcal{H}) = \mathbb{E}_S[R_S(\ell \circ \mathcal{H})]$. Since $\hat{h} \in \mathcal{H}$, theorem 4.9 gives

$$\mathbb{E}_S\big[L_D(\hat{h}) - L_S(\hat{h})\big] \le \mathbb{E}_S\Big[\sup_{h \in \mathcal{H}} \big(L_D(h) - L_S(h)\big) \Big] \le 2R_{D,m}(\ell \circ \mathcal{H}) \le \sqrt{\frac{2\ln \tau_{\mathcal{H}}(m)}{m}}.$$

The second bound follows in the same way from theorem 4.10. □

Thresholds and decision stumps. A **threshold function** on $\mathcal{X} = \mathbb{R}$ is $h_t(x) = -1$ for $x \le t$ and $h_t(x) = +1$ for $x > t$, with $t \in \mathbb{R}$; let $\mathcal{H}_{\text{thr}} = \{h_t : t \in \mathbb{R}\}$. On points $x_1 < \cdots < x_m$, h_t labels the points $\le t$ with -1 and the others with $+1$, so the possible labellings are $(-1, \dots, -1, +1, \dots, +1)$ with $k = 0, 1, \dots, m$ minus signs, and $\tau_{\mathcal{H}_{\text{thr}}}(m) = m + 1$. For instance $\tau_{\mathcal{H}_{\text{thr}}}(2) = 3$: the labelling $(+1, -1)$ of $x_1 < x_2$ is impossible. The decision stumps of section 4, written as

$$h_{t,b}(x) = \begin{cases} b & \text{if } x \le t, \\ -b & \text{if } x > t, \end{cases} \qquad \mathcal{H}_{\text{ds}} = \big\{ h_{t,b} : t \in \mathbb{R},\ b \in \{\pm 1\} \big\},$$

are the thresholds ($b = -1$) and their negations ($b = +1$), which produce the $m + 1$ labellings $(+1, \dots, +1, -1, \dots, -1)$. The two constant labellings appear in both lists, so $\tau_{\mathcal{H}_{\text{ds}}}(m) = 2(m+1) - 2 = 2m$.

5.2 VC dimension and Sauer's lemma

Shattering. Theorem 5.6 is informative only when $\tau_{\mathcal{H}}(m)$ is much smaller than the trivial bound 2^m. For the class of all classifiers $\mathcal{H} = \{\pm 1\}^{\mathcal{X}}$, $\tau_{\mathcal{H}}(m) = 2^m$ for every m, and ERM over it generalises poorly. Instead $\tau_{\mathcal{H}_{\text{thr}}}(m) = m + 1 < 2^m$ for $m > 1$ and $\tau_{\mathcal{H}_{\text{ds}}}(m) = 2m < 2^m$ for $m > 2$. The VC dimension is the complexity measure built on this observation.

Definition 5.7 (Shattering, VC dimension)**.** A set $C = \{x_1, \dots, x_m\} \subseteq \mathcal{X}$ is **shattered** by $\mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$ if $|\mathcal{H}_{|C}| = 2^m$, that is, for every labelling $s \in \{\pm 1\}^m$ there is $h_s \in \mathcal{H}$ with $h_s(x_i) = s_i$ for all i. The **Vapnik–Chervonenkis (VC) dimension** of a non-empty $\mathcal{H}$ is the size of the largest set shattered by $\mathcal{H}$,

$$\operatorname{VCdim}(\mathcal{H}) = \max\{m \ge 0 : \tau_{\mathcal{H}}(m) = 2^m\},$$

and $\operatorname{VCdim}(\mathcal{H}) = \infty$ if $\mathcal{H}$ shatters arbitrarily large sets.

Every subset of a shattered set is shattered (restrict the labellings to it). Hence $\operatorname{VCdim}(\mathcal{H}) = d < \infty$ exactly when some set of size d is shattered by $\mathcal{H}$ (other sets of size d need not be) and no set of size $d + 1$ is.

Decision stumps. $\operatorname{VCdim}(\mathcal{H}_{\text{ds}}) = 2$. From the growth function: $\tau_{\mathcal{H}_{\text{ds}}}(2) = 4 = 2^2$, while $\tau_{\mathcal{H}_{\text{ds}}}(m) = 2m < 2^m$ for $m \ge 3$. Directly: two points $x_1 < x_2$ are shattered, since the four labellings are produced by

$(-1,-1)$	$(-1,+1)$	$(+1,-1)$	$(+1,+1)$
$b = -1,\ t \ge x_2$	$b = -1,\ x_1 \le t < x_2$	$b = +1,\ x_1 \le t < x_2$	$b = +1,\ t \ge x_2$

For three points $x_1 < x_2 < x_3$ the labellings $(-1, +1, -1)$ and $(+1, -1, +1)$ are impossible, because a stump changes value only once along the line; so no set of size 3 is shattered.

Axis-parallel rectangles. Let $\mathcal{X} = \mathbb{R}^2$ and let $\mathcal{H}$ consist of the indicators of axis-parallel rectangles,

$$h_{a,b,c,d}(x) = \begin{cases} +1 & \text{if } a \le x^{(1)} \le b \text{ and } c \le x^{(2)} \le d, \\ -1 & \text{otherwise.} \end{cases}$$

Then $\operatorname{VCdim}(\mathcal{H}) = 4$. The four points $(\pm 1, 0)$, $(0, \pm 1)$ are shattered (fig. 5(a)): for each subset, the smallest rectangle containing it contains none of the other points, and the empty subset is selected by a rectangle far from all points. No set of 5 points is shattered: take a point with smallest and one with largest first coordinate, and a point with smallest and one with largest second coordinate. Any rectangle containing these (at most 4) points contains all 5 (fig. 5(b)), so labelling them $+1$ and a remaining point -1 is impossible.

Convex polygons. For a convex polygon $P \subseteq \mathbb{R}^2$ let $h_P(x) = +1$ if $x \in P$ and $h_P(x) = -1$ otherwise, and $\mathcal{H} = \{h_P : P \text{ convex polygon}\}$. Then $\mathrm{VCdim}(\mathcal{H}) = \infty$: for every m, any m distinct points on a circle are shattered. Given a labelling, let P be the convex hull of the points labelled $+1$. It is a polygon inscribed in the circle, so it meets the circle only at its vertices, and the points labelled -1 lie outside it (fig. 5(c)).

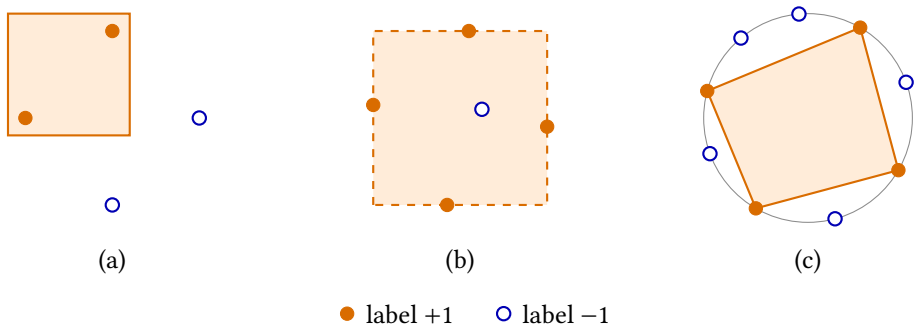


Figure 5: Shattering by axis-parallel rectangles and by convex polygons.

Sauer's lemma.

Theorem 5.8 (Sauer's lemma). *Let $\mathcal{H} \subseteq \{\pm 1\}^{\mathcal{X}}$ be non-empty with $\mathrm{VCdim}(\mathcal{H}) = d < \infty$. For all $m \in \mathbb{N}$,*

$$\tau_{\mathcal{H}}(m) \le \sum_{i=0}^{d} \binom{m}{i},$$

and for all $m \ge d \ge 1$ the simpler bound $\tau_{\mathcal{H}}(m) \le \left(\frac{em}{d}\right)^d$ holds.

So $\tau_{\mathcal{H}}(m) = O(m^d)$ grows polynomially in m, not as 2^m.

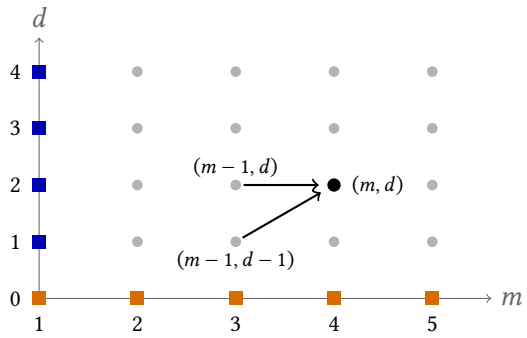


Figure 6: Induction scheme in the proof of Sauer's lemma.

Proof. We prove the first bound by induction on m and d, for every non-empty class of VC dimension at most d; since the bound increases with d, this is the same statement. The two base cases cover the bottom row and the left column of the grid of pairs (m, d) in fig. 6, and the inductive step obtains (m, d) from $(m-1, d)$ and $(m-1, d-1)$.

Base case $d = 0$, $m \ge 1$. No single point is shattered, so all $h \in \mathcal{H}$ take the same value at every x: $|\mathcal{H}| = 1$ and $\tau_{\mathcal{H}}(m) = 1 = \binom{m}{0}$.

Base case $m = 1$, $d \ge 1$. $\tau_{\mathcal{H}}(1) \le 2 = \binom{1}{0} + \binom{1}{1} = \sum_{i=0}^{d} \binom{1}{i}$.

Inductive step, $m \ge 2$, $d \ge 1$. Assume the bound for $(m-1, d)$ and $(m-1, d-1)$. Let $C = (x_1, \dots, x_m)$ consist of distinct points and $C' = (x_2, \dots, x_m)$. Every $(y_2, \dots, y_m) \in \mathcal{H}_{|C'}$ extends to one or two vectors of $\mathcal{H}_{|C}$, namely $(-1, y_2, \dots, y_m)$ and/or $(+1, y_2, \dots, y_m)$. Let

$$Y = \left\{ (y_2, \dots, y_m) \in \mathcal{H}_{|C'} : (-1, y_2, \dots, y_m) \in \mathcal{H}_{|C} \text{ and } (+1, y_2, \dots, y_m) \in \mathcal{H}_{|C} \right\}.$$

Counting each element of $\mathcal{H}_{|C'}$ once, and once more if it is in Y, gives $|\mathcal{H}_{|C}| = |\mathcal{H}_{|C'}| + |Y|$. We bound both terms.

First, $|\mathcal{H}_{|C'}| \le \tau_{\mathcal{H}}(m-1) \le \sum_{i=0}^{d} \binom{m-1}{i}$ by the induction hypothesis for $(m-1, d)$.

Second, view $Y \subseteq \{\pm 1\}^{C'}$ as a hypothesis class on C'. If Y shattered a set $C'' \subseteq C'$ of size d, then $\mathcal{H}$ would shatter $C'' \cup \{x_1\}$, of size $d+1$: for any labelling of $C'' \cup \{x_1\}$, some vector of Y realises it on C'', and both of its extensions are in $\mathcal{H}_{|C}$, so one of them also gives x_1 the required label. This contradicts $\mathrm{VCdim}(\mathcal{H}) \le d$. Hence $\mathrm{VCdim}(Y) \le d-1$ and, by the induction hypothesis for $(m-1, d-1)$, $|Y| \le \sum_{i=0}^{d-1} \binom{m-1}{i}$ (trivially if $Y = \emptyset$).

Adding the two bounds, shifting the index of the second sum and using Pascal's rule $\binom{m-1}{i} + \binom{m-1}{i-1} = \binom{m}{i}$ together with $\binom{m-1}{0} = 1 = \binom{m}{0}$,

$$|\mathcal{H}_{|C}| \le \sum_{i=0}^{d} \binom{m-1}{i} + \sum_{i=1}^{d} \binom{m-1}{i-1} = \binom{m-1}{0} + \sum_{i=1}^{d} \left(\binom{m-1}{i} + \binom{m-1}{i-1} \right) = \sum_{i=0}^{d} \binom{m}{i}.$$

This holds for every C of size m, hence for $\tau_{\mathcal{H}}(m)$.

Simpler bound. Let $m \ge d \ge 1$. Multiplying the i-th term by $(\frac{m}{d})^{d-i} \ge 1$, then adding the terms $i = d+1, \dots, m$ (all ≥ 0),

$$\begin{aligned} \tau_{\mathcal{H}}(m) \le \sum_{i=0}^{d} \binom{m}{i} \left(\frac{m}{d}\right)^{d-i} &= \left(\frac{m}{d}\right)^{d} \sum_{i=0}^{d} \binom{m}{i} \left(\frac{d}{m}\right)^{i} \le \left(\frac{m}{d}\right)^{d} \sum_{i=0}^{m} \binom{m}{i} \left(\frac{d}{m}\right)^{i} \\ &= \left(\frac{m}{d}\right)^{d} \left(1 + \frac{d}{m}\right)^{m} \le \left(\frac{m}{d}\right)^{d} e^{d} = \left(\frac{em}{d}\right)^{d}, \end{aligned}$$

by the binomial theorem and $(1 + \frac{x}{n})^n \le e^x$. □

5.3 VC dimension of common hypothesis classes

Finite classes. If $\mathcal{H}$ is finite, then $\mathrm{VCdim}(\mathcal{H}) \le \log_2 |\mathcal{H}|$: since $\tau_{\mathcal{H}}(m) \le |\mathcal{H}|$ for every m, a shattered set of size d gives $2^d = \tau_{\mathcal{H}}(d) \le |\mathcal{H}|$.

Linear classifiers. The **linear classifiers** on $\mathcal{X} = \mathbb{R}^p$ are

$$\mathcal{H}_{\mathrm{lin}} = \left\{ x \mapsto \mathrm{sign}(\langle w, x \rangle + b) \,:\, w \in \mathbb{R}^p,\ b \in \mathbb{R} \right\}, \qquad \langle w, x \rangle = w^{\top} x;$$

ERM over $\mathcal{H}_{\mathrm{lin}}$ is related to support vector machines and the perceptron.

Theorem 5.9 (VC dimension of linear classifiers)**.** $\mathrm{VCdim}(\mathcal{H}_{\mathrm{lin}}) = p + 1$.

Proof. A set of $p+1$ points is shattered. Take $\{e_1, \dots, e_p, 0\}$, with e_i the i-th standard basis vector. Given labels s_i for e_i and s_0 for 0, let $w = (s_1, \dots, s_p)$ and $b = s_0/2$. Then $\langle w, 0 \rangle + b = s_0/2$ has sign s_0, and $\langle w, e_i \rangle + b = s_i + s_0/2$ has sign s_i because $|s_i| = 1 > \frac{1}{2}$.

No set of $p+2$ points is shattered. Suppose $x_1, \dots, x_{p+2} \in \mathbb{R}^p$ are shattered. The homogeneous linear system

$$\begin{pmatrix} x_1 & x_2 & \cdots & x_{p+2} \\ 1 & 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_{p+2} \end{pmatrix} = 0$$

has $p+1$ equations and $p+2$ unknowns, so it has a solution $a \neq 0$. Let $I_+ = \{i : a_i > 0\}$ and $I_- = \{i : a_i < 0\}$; both are non-empty, because $a \neq 0$ and $\sum_i a_i = 0$. The system says $\sum_i a_i x_i = 0$ and $\sum_i a_i = 0$; dropping the indices with $a_i = 0$ and moving those in I_- to the right side,

$$\sum_{i \in I_+} a_i = \sum_{i \in I_-} |a_i|, \qquad \sum_{i \in I_+} a_i x_i = \sum_{i \in I_-} |a_i| x_i.$$

Since the points are shattered, some (w, b) gives $\langle w, x_i\rangle + b > 0$ for $i \in I_+$ and $\langle w, x_i\rangle + b < 0$ for $i \in I_-$. Then

$$0 < \sum_{i\in I_+} a_i\big(\langle w, x_i\rangle + b\big) = \Big\langle w, \sum_{i\in I_+} a_i x_i \Big\rangle + b \sum_{i\in I_+} a_i$$
$$= \Big\langle w, \sum_{i\in I_-} |a_i| x_i \Big\rangle + b \sum_{i\in I_-} |a_i| = \sum_{i\in I_-} |a_i|\big(\langle w, x_i\rangle + b\big) < 0,$$

a contradiction. □

Figure 7 shows both parts of the proof for $p = 2$.

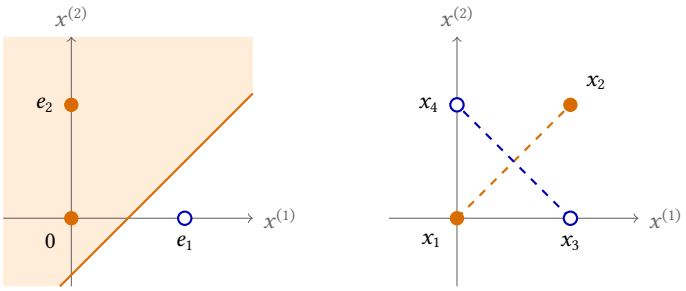


Figure 7: Labellings by linear classifiers in $\mathbb{R}^2$.

For ERM over $\mathcal{H}_{\text{lin}}$ the following bound holds with probability at least $1 - \delta$ (stated without proof):

$$L_D(\hat{h}) \le L_D(\mathcal{H}_{\text{lin}}) + 2\sqrt{\frac{8}{m}\Big((p+1)\ln\frac{2em}{p+1} + \ln\frac{4}{\delta}\Big)} \le L_D(\mathcal{H}_{\text{lin}}) + O\Big(\sqrt{\frac{p\ln m}{m}}\Big).$$

Kernel classifiers. The **homogeneous linear classifiers** $\mathcal{H}_{\text{lin-h}} = \{x \mapsto \operatorname{sign}(\langle w, x\rangle) : w \in \mathbb{R}^p\}$, whose decision boundaries pass through the origin, have $\operatorname{VCdim}(\mathcal{H}_{\text{lin-h}}) = p$. Every symmetric positive semidefinite kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ comes with a **reproducing kernel Hilbert space** (RKHS) $\mathcal{K}$, with inner product $\langle\cdot,\cdot\rangle_{\mathcal{K}}$, and a map $\phi : \mathcal{X} \to \mathcal{K}$, $x \mapsto \phi_x$, such that $k(x, x') = \langle \phi_x, \phi_{x'}\rangle_{\mathcal{K}}$. A **kernel classifier** is a homogeneous linear classifier in the RKHS,

$$\mathcal{H}_{\mathcal{K}} = \big\{ h_w(x) = \operatorname{sign}(\langle w, \phi_x\rangle_{\mathcal{K}}) : w \in \mathcal{K} \big\},$$

and $\operatorname{VCdim}(\mathcal{H}_{\mathcal{K}}) = \dim \mathcal{K}$, which may be infinite: the RBF kernel $k(x, x') = e^{-\|x-x'\|^2}$ has an infinite-dimensional $\mathcal{K}$.

1-nearest neighbour. The 1-NN predictor of section 3, $h_S(x) = y_{\pi_1(x)}$ with $x_{\pi_1(x)}$ the nearest neighbour of x in S, ranges over the class

$$\mathcal{H}_{\text{1-NN}} = \big\{ h_S : S \in (\mathcal{X} \times \mathcal{Y})^m,\ m \in \mathbb{N} \big\}.$$

Its VC dimension is infinite: to shatter any $C = \{x_1, \dots, x_m\}$, use $h_{S_1}, \dots, h_{S_{2^m}}$, where the samples $S_1, \dots, S_{2^m}$ have features $x_1, \dots, x_m$ and carry the 2^m labellings. Each x_i is its own nearest neighbour in S_j, so h_{S_j} reproduces on C the labels of S_j. Hence 1-NN has no uniform convergence at finite m.

Two-layer neural networks. Consider the simplified network of fig. 8: the input is $x \in \mathbb{R}^p$, the hidden layer has N units

$$f_i(x) = \operatorname{sign}(\langle w_i, x\rangle + b_i), \qquad i = 1, \dots, N,$$

collected in $f(x) = (f_1(x), \dots, f_N(x)) \in \{\pm 1\}^N$, and the output is $h(x) = \operatorname{sign}(\langle w, f(x)\rangle + b)$. Let $\mathcal{H}$ be the class of all such h, parametrised by $w \in \mathbb{R}^N$, $w_1, \dots, w_N \in \mathbb{R}^p$ and $b, b_1, \dots, b_N \in \mathbb{R}$.

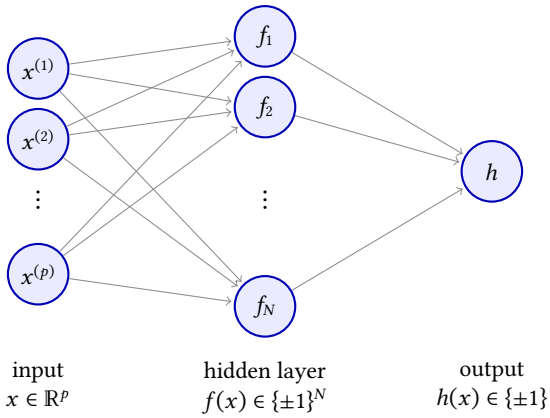


Figure 8: Simplified two-layer network.

The network combines $N+1$ linear classifiers, so its growth function is bounded through the growth functions of combined classes. For a class $\mathcal{G}$ of functions from $\mathcal{X}$ to any set, $\tau_{\mathcal{G}}(m)$ is defined as in definition 5.5: the largest number of distinct vectors $(g(x_1), \dots, g(x_m))$, $g \in \mathcal{G}$, over sets $\{x_1, \dots, x_m\} \subseteq \mathcal{X}$.

Lemma 5.10 (Concatenating classifiers). *Let $\mathcal{G}' \subseteq \mathcal{Y}'^{\mathcal{X}}$ and $\mathcal{G}'' \subseteq \mathcal{Y}''^{\mathcal{X}}$, and*

$$\mathcal{G} = \mathcal{G}' \times \mathcal{G}'' = \left\{ x \mapsto (g'(x), g''(x)) \,:\, g' \in \mathcal{G}',\ g'' \in \mathcal{G}'' \right\} \subseteq (\mathcal{Y}' \times \mathcal{Y}'')^{\mathcal{X}}.$$

Then $\tau_{\mathcal{G}}(m) \le \tau_{\mathcal{G}'}(m)\, \tau_{\mathcal{G}''}(m)$.

Lemma 5.11 (Composition of classifiers). *Let $\mathcal{G}' \subseteq \mathcal{Y}^{\mathcal{X}}$ and $\mathcal{G}'' \subseteq \mathcal{Z}^{\mathcal{Y}}$, and*

$$\mathcal{G} = \mathcal{G}'' \circ \mathcal{G}' = \left\{ g'' \circ g' \,:\, g' \in \mathcal{G}',\ g'' \in \mathcal{G}'' \right\} \subseteq \mathcal{Z}^{\mathcal{X}}.$$

Then $\tau_{\mathcal{G}}(m) \le \tau_{\mathcal{G}'}(m)\, \tau_{\mathcal{G}''}(m)$.

Theorem 5.12 (VC dimension of a two-layer network). $\operatorname{VCdim}(\mathcal{H}) = O\big(pN \log_2(pN)\big)$.

Proof. Write $\mathcal{H} = \mathcal{H}' \circ (\mathcal{H}_1 \times \cdots \times \mathcal{H}_N)$, where $\mathcal{H}_i \subseteq \{\pm 1\}^{\mathbb{R}^p}$ is the class of the i-th hidden unit and $\mathcal{H}' \subseteq \{\pm 1\}^{\mathbb{R}^N}$ that of the output unit. All are classes of linear classifiers, so $\operatorname{VCdim}(\mathcal{H}_i) = p+1$ and $\operatorname{VCdim}(\mathcal{H}') = N+1$ by theorem 5.9, and Sauer's lemma gives

$$\tau_{\mathcal{H}_i}(m) \le \left(\frac{em}{p+1}\right)^{p+1} < (me)^{p+1} \quad (m \ge p+1), \qquad \tau_{\mathcal{H}'}(m) \le \left(\frac{em}{N+1}\right)^{N+1} < (me)^{N+1} \quad (m \ge N+1).$$

By lemma 5.10 (applied $N-1$ times) and lemma 5.11, for $m \ge \max\{p, N\} + 1$ and $m > e$,

$$\tau_{\mathcal{H}}(m) \le \tau_{\mathcal{H}'}(m)\, \tau_{\mathcal{H}_1}(m) \cdots \tau_{\mathcal{H}_N}(m) < (me)^{N(p+1)+N+1} < m^{8pN},$$

where the last step uses $me < m^2$ (as $m > e$) and $N(p+1) + N + 1 = Np + 2N + 1 \le 4Np$ (as $N \le Np$ and $1 \le Np$), so that $(me)^{Np+2N+1} < m^{2(Np+2N+1)} \le m^{8Np}$.

If $\tau_{\mathcal{H}}(m) < 2^m$, no set of size m is shattered, so $\operatorname{VCdim}(\mathcal{H}) < m$ (a shattered set of size $\ge m$ would have shattered subsets of size m). To find such an m we use the following fact: for $c > 0$, $x > \max\{2, 3c\}$ and $m = 3cx \log_2 x$,

$$2^m > m^{cx}.$$

Indeed $m > 0$ (as $x > 2$), and taking $\log_2$ and dividing by $cx > 0$,

$$2^m > m^{cx} \iff \frac{m}{cx} > \log_2 m \iff 3\log_2 x > \log_2(3cx \log_2 x) \iff x^3 > 3cx \log_2 x \iff x^2 > 3c \log_2 x,$$

which holds because $x^2 > 3cx$ (as $x > 3c$) and $x > \log_2 x$.

Case $Np > 24$. Apply the fact with $c = 8$ and $x = Np > 24 = \max\{2, 3c\}$: for $m = 24Np\log_2(Np)$, which satisfies $m > e$ and $m \geq \max\{p, N\} + 1$, we get $\tau_{\mathcal{H}}(m) < m^{8pN} < 2^m$, hence $\mathrm{VCdim}(\mathcal{H}) < 24Np\log_2(Np)$.

Case $Np \leq 24$. Then $\tau_{\mathcal{H}}(m) < m^{8pN} \leq m^{8\cdot 24} < 2^m$ for every m larger than an absolute constant, so $\mathrm{VCdim}(\mathcal{H}) = O(1)$.

In both cases $\mathrm{VCdim}(\mathcal{H}) = O(pN\log_2(pN))$. □

The corresponding bound for ERM over $\mathcal{H}$, stated without proof, holds with probability at least $1 - \delta$:

$$L_D(\hat{h}) \leq L_D(\mathcal{H}) + O\left(\sqrt{\frac{pN\ln(pN)\ln m + \ln\frac{1}{\delta}}{m}}\right).$$